Hypothesis Testing

Class Objectives

- Developing Null and Alternative Hypotheses
- Type I and Type II Errors- Explanation
- Population Mean: Sigma Known
- Population Mean: Sigma Unknown
- Population Proportion

Hypothesis Testing

- Hypothesis testing can be used to determine whether a statement about the value of a population parameter should or should not be rejected.
- The null hypothesis, denoted by H_0 , is a tentative assumption about a population parameter
- The alternative hypothesis, denoted by Ha, is the opposite of what is stated in the null hypothesis
- The hypothesis testing procedure uses data from a sample to test the two competing statements indicated by H0 and Ha.

Developing Null and Alternative Hypotheses

- It is not always obvious how the null and alternative hypotheses should be formulated
- Care must be taken to structure the hypotheses appropriately so that the test conclusion provides the information the researcher wants
- The context of the situation is very important in determining how the hypotheses should be stated
- In some cases it is easier to identify the alternative hypothesis first. In other cases the null is easier
- Correct hypothesis formulation will take practice

Developing Null and Alternative Hypotheses

Alternative Hypothesis as a Research Hypothesis

- Many applications of hypothesis testing involve an attempt to gather evidence in support of a research hypothesis
- In such cases, it is often best to begin with the alternative hypothesis and make it the conclusion that the researcher hopes to support
- The conclusion that the research hypothesis is true is made if the sample data provide sufficient evidence to show that the null hypothesis can be rejected

Developing Null and Alternative Hypotheses

Alternative Hypothesis as a Research Hypothesis

- Example: A new manufacturing method is believed to be better than the current method.
- Alternative Hypothesis:
  - The new manufacturing method is better.
- Null Hypothesis:
  - The new method is no better than the old method.

Developing Null and Alternative Hypotheses

- Alternative Hypothesis as a Research Hypothesis
- Example: A new bonus plan, that is developed in an attempt to increase sales
- Alternative Hypothesis:
  - The new bonus plan increase sales
- Null Hypothesis:
  - The new bonus plan does not increase sales

Developing Null and Alternative Hypotheses

- Alternative Hypothesis as a Research Hypothesis
- Example:
  - A new drug is developed with the goal of lowering Cholesterol-level more than the existing drug
- Alternative Hypothesis:
  - The new drug lowers Cholesterol-level more than the existing drug
- Null Hypothesis:
  - The new drug does not lower Cholesterol-level more than the existing drug

Developing Null and Alternative Hypotheses

- Null Hypothesis as an assumption to be challenged
- We might begin with a belief or assumption that a statement about the value of a population parameter is true
- We then using a hypothesis test to challenge the assumption and determine if there is statistical evidence to conclude that the assumption is incorrect
- In these situations, it is helpful to develop the null hypothesis first

Developing Null and Alternative Hypotheses

- Null Hypothesis as an Assumption to be Challenged
- Example:
  - The label on a milk bottle states that it contains 1000 ml
- Null Hypothesis:
  - The label is correct. $\mu \geq 1000$ ml
- Alternative Hypothesis:
  - The label is incorrect. $\mu < 1000$ ml

Null and Alternative Hypotheses about a Population Mean μ

- The equality part of the hypotheses always appears in the null hypothesis
- In general, a hypothesis test about the value of a population mean μ must take one of the following three forms (where μ_0 is the hypothesized value of the population mean)

$H_0: \mu \geq \mu_0$	$H_0: \mu \leq \mu_0$	$H_0: \mu = \mu_0$
$H_a: \mu < \mu_0$	$H_a: \mu > \mu_0$	$H_a: \mu \neq \mu_0$
One-tailed (lower-tail)	One-tailed (upper-tail)	Two-tailed

Null and Alternative Hypotheses

- A major hospital in Chennai provides one of the most comprehensive emergency medical services in the world
- Operating in a multiple hospital system with approximately 10 mobile medical units, the service goal is to respond to medical emergencies with a mean time of 8 minutes or less
- The director of medical services wants to formulate a hypothesis test that could use a sample of emergency response times to determine whether or not the service goal of 8 minutes or less is being achieved.

Null and Alternative Hypotheses

H_0: $\mu \leq 8$

The emergency service is meeting the response goal; no follow-up action is necessary.

H_a: $\mu > 8$

The emergency service is not meeting the response goal; appropriate follow-up action is necessary.

where: μ = mean response time for the population of medical emergency requests

Type I Error

- Because hypothesis tests are based on sample data, we must allow for the possibility of errors
- A Type I error is rejecting H_0 when it is true
- The probability of making a Type I error when the null hypothesis is called the level of significance
- Applications of hypothesis testing that only control the Type I error are often called significance tests

Type II Error

- A Type II error is accepting H_0 when it is false.
- It is difficult to control for the probability of making a Type II error.
- Statisticians avoid the risk of making a Type II error by using "do not reject H_0" and not "accept H_0".

Type I and Type II Errors

	Population Condition	
Conclusion	H0 True ($\mu \leq 8$)	H0 False ($\mu > 8$)
Accept H0 (Conclude $\mu \leq 8$)	Correct Decision	Type II Error
Reject H0 (Conclude $\mu > 8$)	Type I Error	Correct Decision

Three Approaches for Hypothesis Testing

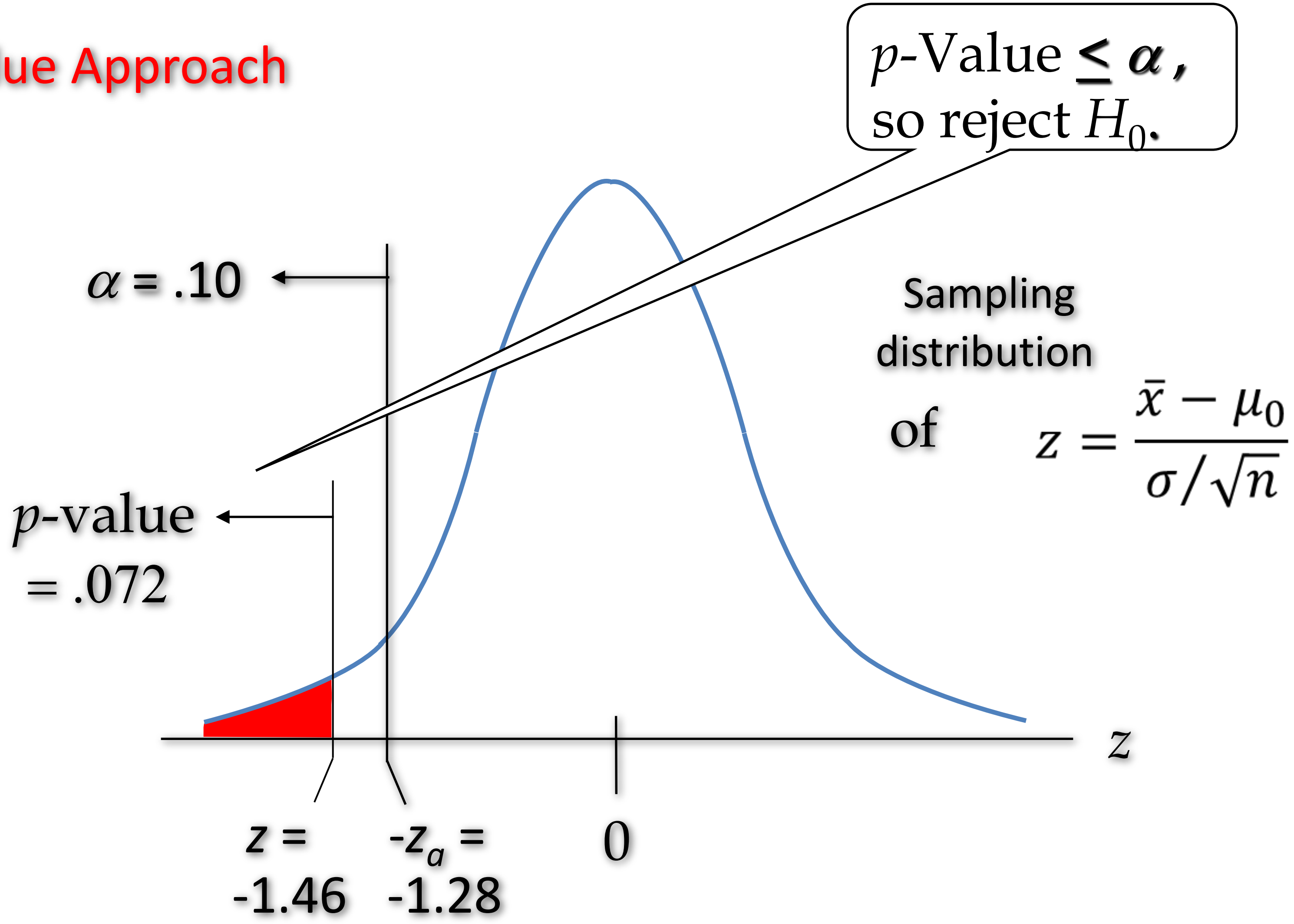
- P- Value
- Critical Value
- Confidence Interval Value

p-Value Approach to One-Tailed Hypothesis Testing

- The *p*-value is the probability, computed using the test statistic, that measures the support (or lack of support) provided by the sample for the null hypothesis
- If the *p*-value is less than or equal to the level of significance α, the value of the test statistic is in the rejection region
- Reject H_0 if the *p*-value $\leq \alpha$

Lower-Tailed Test About a Population Mean: σ Known

p-Value Approach



p-Value Approach

Finding P Value

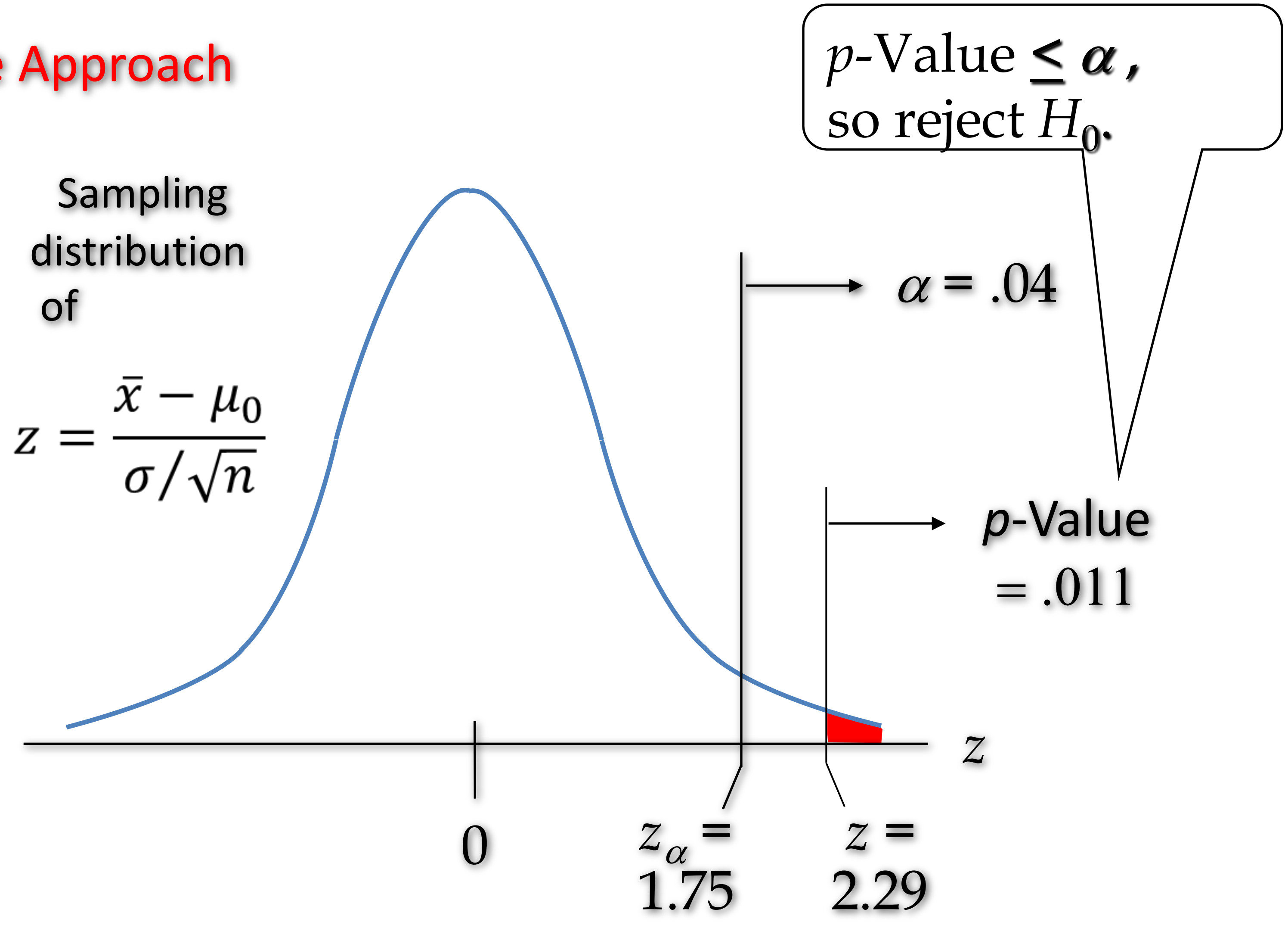
```
In [3]: stats.norm.cdf(-1.46)
Out[3]: 0.07214503696589378
```

Finding Z Value

```
In [5]: stats.norm.ppf(0.1)
Out[5]: -1.2815515655446004
```

Upper-Tailed Test About a Population Mean : σ Known

p-Value Approach



p-Value Approach

```
In [4]: 1-stats.norm.cdf(1.75)

Out[4]: 0.040059156863817114
```

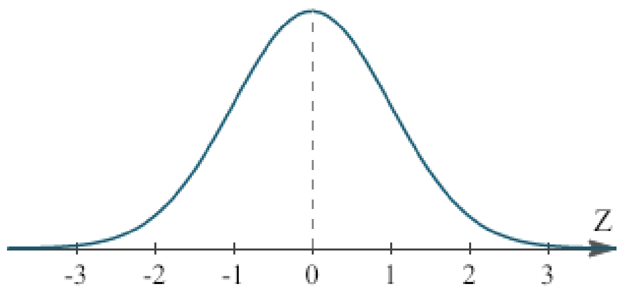
```
In [5]: 1-stats.norm.cdf(2.29)

Out[5]: 0.011010658324411393
```

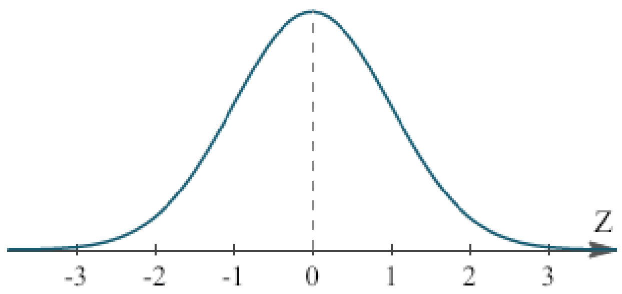
Critical Value Approach to One-Tailed Hypothesis Testing

- The test statistic z has a standard normal probability distribution.
- We can use the standard normal probability distribution table to find the z-value with an area of α in the lower (or upper) tail of the distribution.
- The value of the test statistic that established the boundary of the rejection region is called the critical value for the test.
- The rejection rule is:

  Lower tail: Reject H_0 if $z \le -z_\alpha$

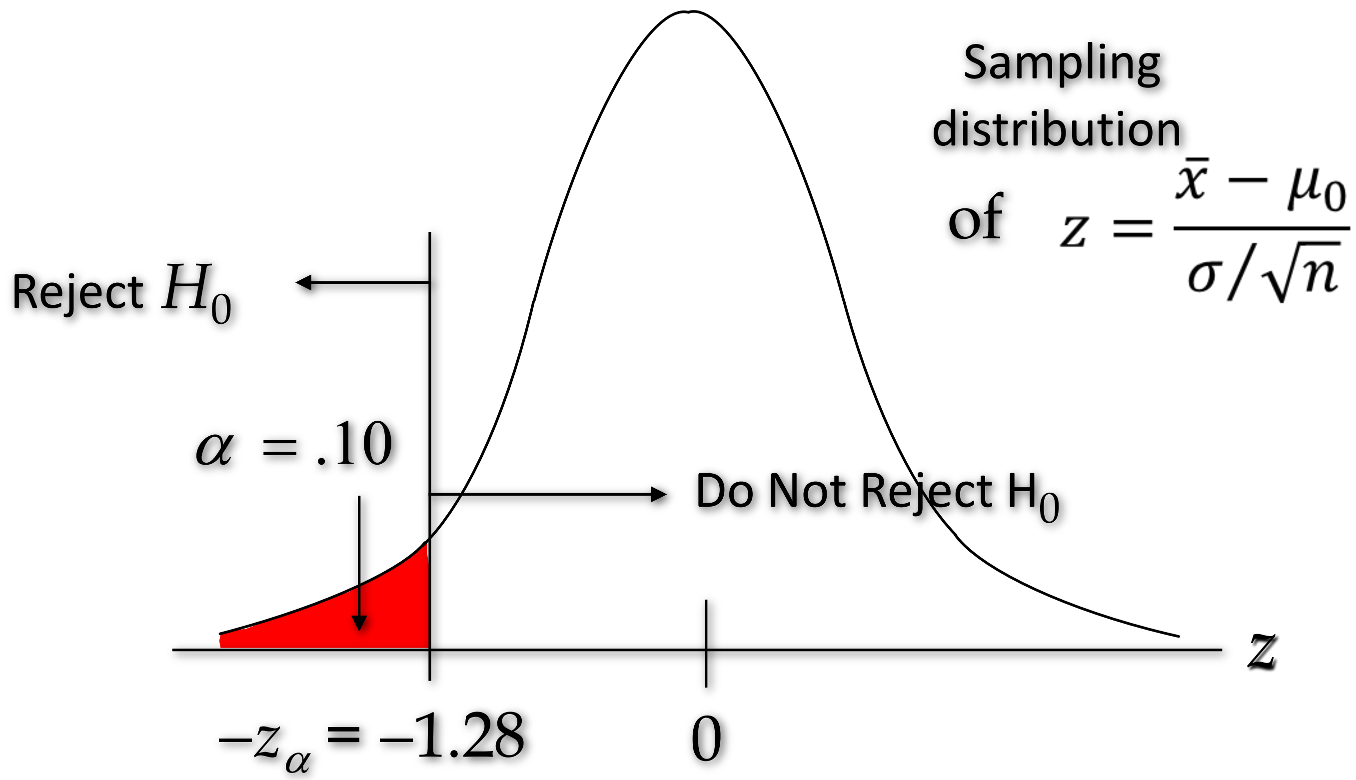
  Upper tail: Reject H_0 if $z \ge z_\alpha$





Lower-Tailed Test About a Population Mean: σ Known

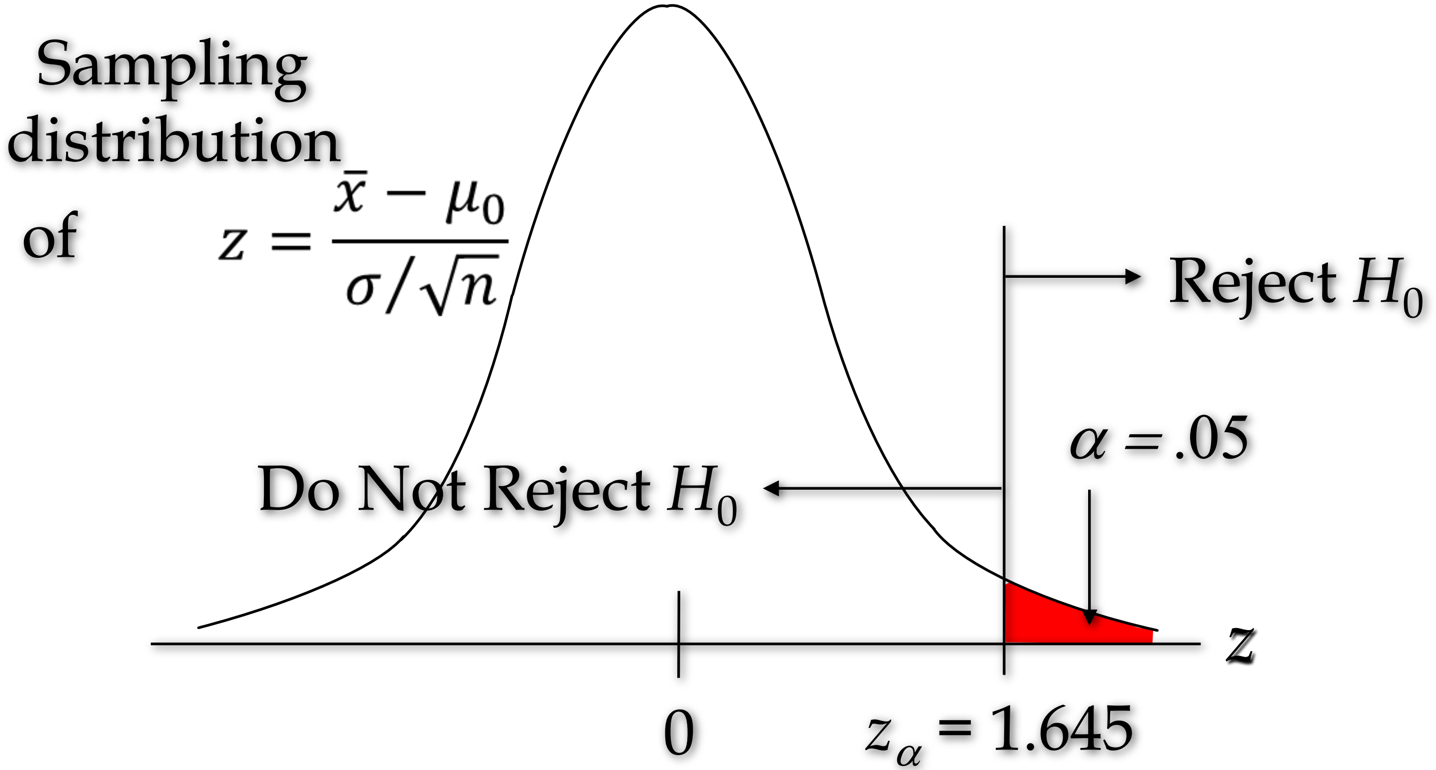
Critical Value Approach



```
In [6]: stats.norm.ppf(0.1)

Out[6]: -1.2815515655446004
```

Upper-Tailed Test About a Population Mean: σ Known

Critical Value Approach



```
In [7]: stats.norm.ppf(0.95)

Out[7]: 1.6448536269514722
```

Steps of Hypothesis Testing – P value approach

- Step 1. Develop the null and alternative hypotheses.
- Step 2. Specify the level of significance α.
- Step 3. Collect the sample data and compute the test statistic.
- <u>*p*-Value Approach</u>
- Step 4. Use the value of the test statistic to compute the *p*-value.
- Step 5. Reject H_0 if *p*-value $\leq \alpha$.

Steps of Hypothesis Testing

Critical Value Approach

•Step 4. Use the level of significance α to determine the critical value and the rejection rule.

•Step 5. Use the value of the test statistic and the rejection rule to determine whether to reject H_0.

Hypothesis Testing

Class Objectives

- Population Mean: Sigma Known –Example

One-Tailed Tests About a Population Mean: σ Known

- Example: The mean response times for a random sample of 30 Pizza Deliveries is 32 minutes
- The population standard deviation is believed to be 10 minutes.
- The pizza delivery services director wants to perform a hypothesis test, with α =0.05 level of significance, to determine whether the service goal of 30 minutes or less is being achieved.

Given Values

- Sample
- Sample mean = 32 Min
- Sample size = 30

- Population
- α =0.05
- Population mean = 30 Min

p -Value Approach

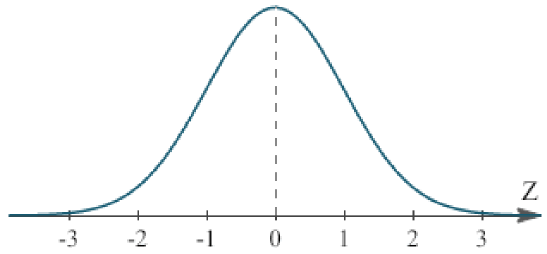
One-Tailed Tests About a Population Mean: σ Known

1. Develop the hypotheses.
2. Specify the level of significance.
3. Compute the value of the test statistic.

$$H_0: \ \mu \le 30$$
$$H_a: \ \mu > 30$$

$$\alpha = .05$$

$$z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}} = \frac{32 - 30}{10 / \sqrt{30}} = 1.09$$



In [8]:
```
1-stats.norm.cdf(1.09)
```
Out[8]:
```
0.13785657203203552
```

One-Tailed Tests About a Population Mean: σ Known

p –Value Approach

4. Compute the p –value.

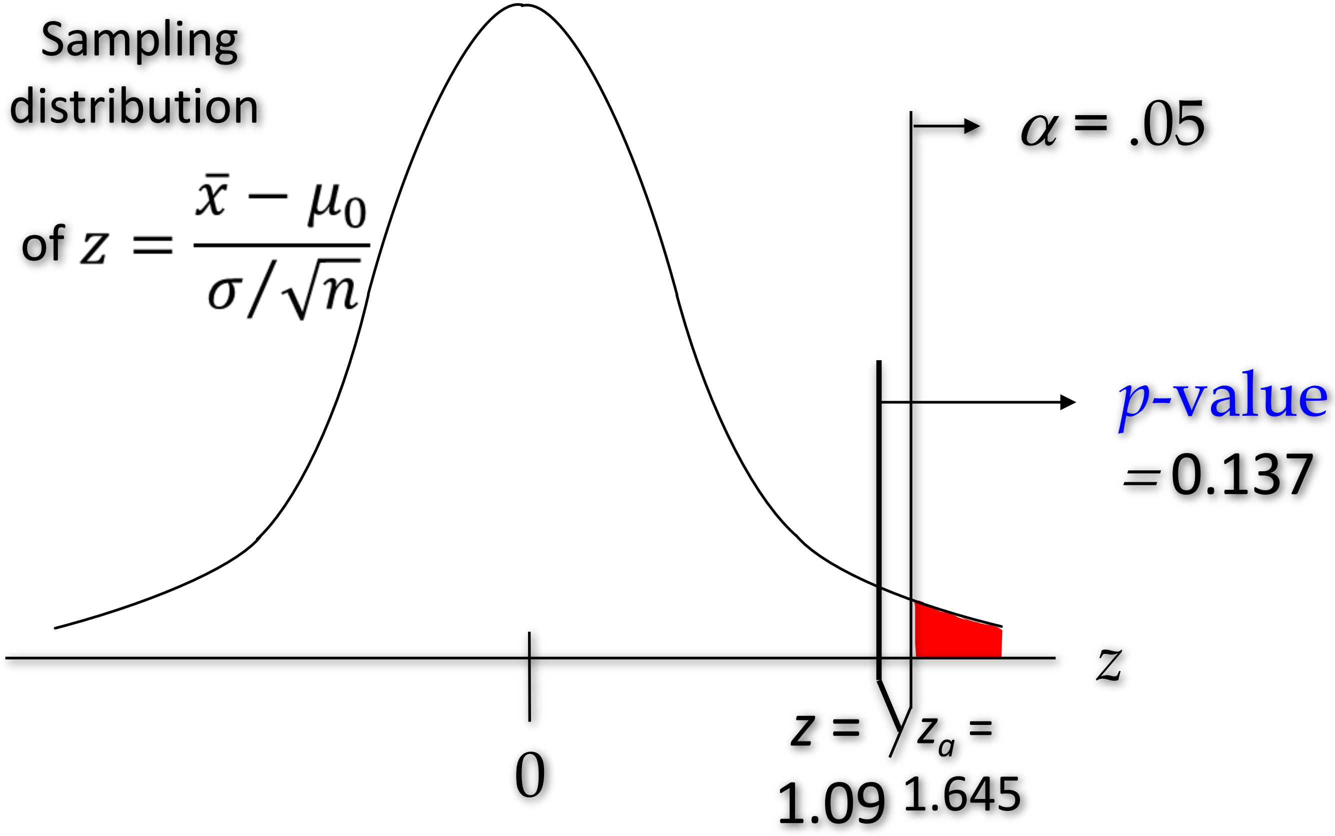
   For z = 1.09, p–value = = 0.137

5. Determine whether to reject H0.

- Because p–value = 0.137 > $\alpha = .05$, we do not reject H0.
- There are not sufficient statistical evidence to infer that Pizza delivery services is not meeting the response goal of 30 minutes.

One-Tailed Tests About a Population Mean: σ Known

p –Value Approach



Critical Value Approach

One-Tailed Tests About a Population Mean: σ Known

Critical Value Approach

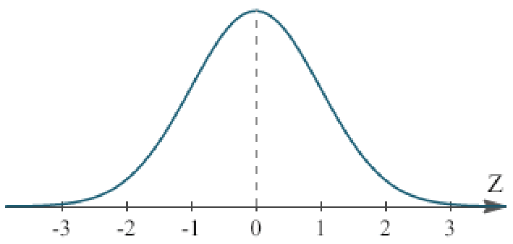
4. Determine the critical value and rejection rule.

   – For $\alpha = .05$, $z_{.05} = 1.645$

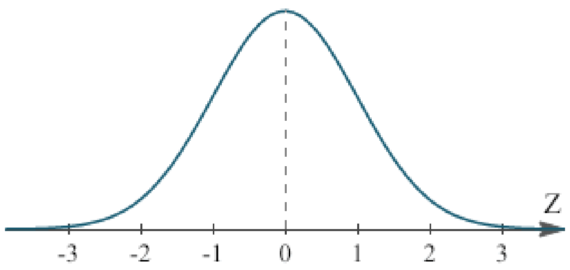
   – Reject H_0 if $z \geq 1.645$

5. Determine whether to reject H_0.

   – Because $1.645 \geq 1.05$, we do not reject H_0.



p-Value Approach to Two-Tailed Hypothesis Testing



Compute the p-value using the following three steps:

1. Compute the value of the test statistic z.
2. If z is in the upper tail $(z > 0)$, find the area under the standard normal curve to the right of z.
3. If z is in the lower tail $(z < 0)$, find the area under the standard normal curve to the left of z.
4. Double the tail area obtained in step 2 to obtain the p –value.

The rejection rule:

Reject H_0 if the p-value $\leq \alpha$.

Critical Value Approach to Two-Tailed Hypothesis Testing

- The critical values will occur in both the lower and upper tails of the standard normal curve.
- Use the standard normal probability distribution table to find $z_{\alpha/2}$ (the z-value with an area of $\alpha/2$ in the upper tail of the distribution).
- The rejection rule is:

Reject H_0 if $z \leq -z_{\alpha/2}$ **or** $z \geq z_{\alpha/2}$.

Two-Tailed Tests About a Population Mean: σ Known

- Example: Milk Carton
- Assume that a sample of 30 milk carton provides a sample mean of 505 ml.
- The population standard deviation is believed to be 10 ml.
- Perform a hypothesis test, at the 0.03 level of significance, population mean 500 ml and to help determine whether the filling process should continue operating or be stopped and corrected.

Given Values

- Sample
- Sample size = 30
- Sample mean = 505 ml

- Population
- Population mean = 500 ml
- Standard deviation = 10 ml
- Significance level 0.03

p –Value approach

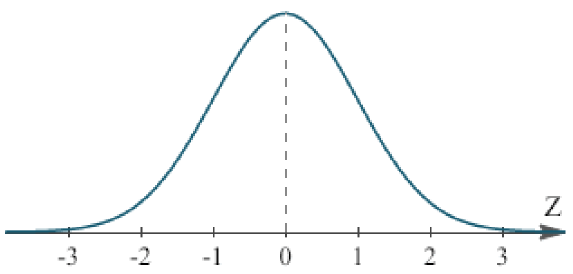
Two-Tailed Tests About a Population Mean: σ Known

1. Determine the hypotheses.
2. Specify the level of significance.
3. Compute the value of the test statistic.

H_0: $\mu = 500$
H_a: $\mu \neq 500$

$\alpha = .03$

$$z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}} = \frac{505 - 500}{10 / \sqrt{30}} = 2.74$$



```
In [9]: 1-stats.norm.cdf(2.74)

Out[9]: 0.003071959218650444
```

```
In [10]: (1-stats.norm.cdf(2.74))*2

Out[10]: 0.006143918437300888
```

Two-Tailed Tests About a Population Mean: σ Known

p –Value Approach

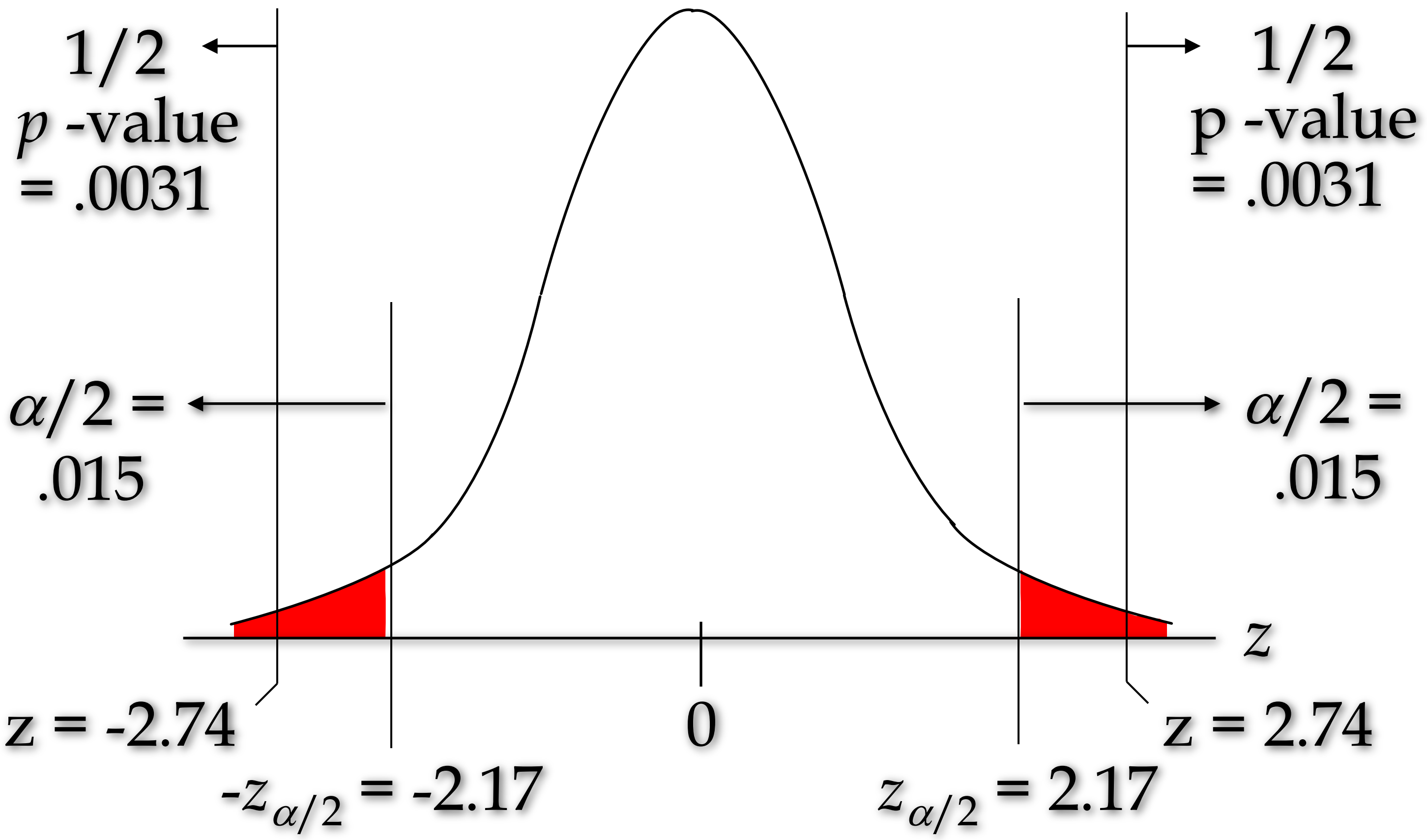
4. Compute the p –value.
   - For $z = 2.74$, p–value = 2(1 - .9969) = .0061

5. Determine whether to reject H0.
   - Because p–value = .0062 < α = .03, we reject H_0.

There is no sufficient statistical evidence to infer that the null hypothesis is true (i.e. the mean filling quantity is not 500 ml)

Two-Tailed Tests About a Population Mean: σ Known

p-Value Approach



Critical Value Approach

Two-Tailed Tests About a Population Mean : σ Known

- **Critical Value Approach**

4. Determine the critical value and rejection rule, for $\alpha/2 = .03/2 = .015$, $z_{.015} = 2.17$

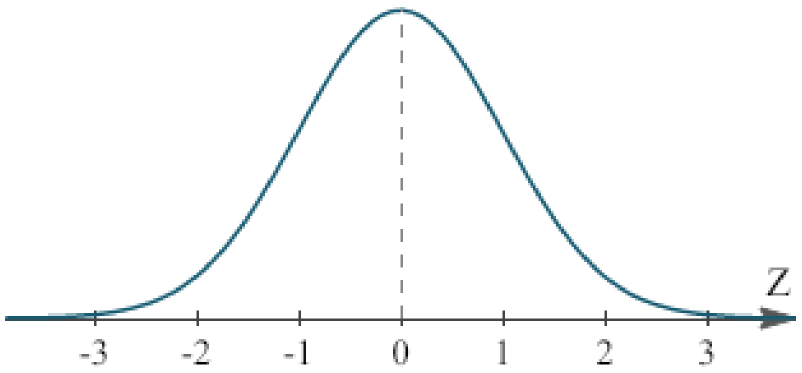
Reject H_0 if $z < -2.17$ or $z > 2.17$

5. Determine whether to reject H_0.

Because 2.74 > 2.17, we reject H0.

There is sufficient statistical evidence to infer that the null hypothesis is not true

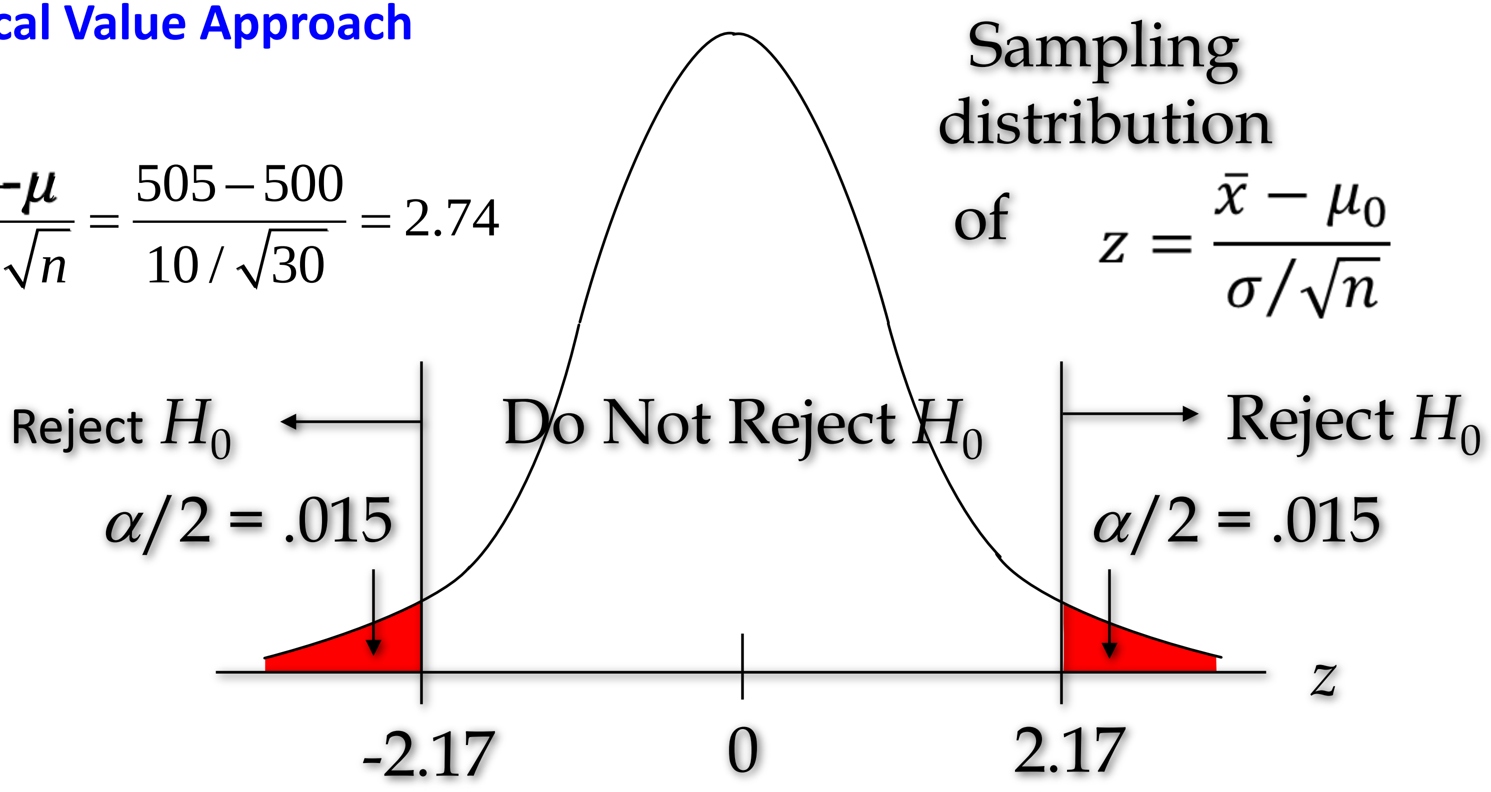
```
In [12]: stats.norm.ppf(0.015)
Out[12]: -2.1700903775845606
```



Two-Tailed Tests About a Population Mean : σ Known

Critical Value Approach

$$z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}} = \frac{505 - 500}{10 / \sqrt{30}} = 2.74$$



Confidence Interval Approach

Confidence Interval Approach to Two-Tailed Tests About a Population Mean

- Select a simple random sample from the population and use the value of the sample mean to develop the confidence interval for the population mean μ.
- If the confidence interval contains the hypothesized value 500, do not reject H_0.
- Otherwise, reject H_0.
- Actually, H_0 should be rejected if μ_0 happens to be equal to one of the end points of the confidence interval.

Confidence Interval Approach to Two-Tailed Tests About a Population Mean

The 97% confidence interval for 500 is

$$\bar{x} \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}} = 505 \pm 2.17\frac{10}{\sqrt{30}} = 505 \pm 3.9619$$

$$= \quad 501.03814\ ,508.96186$$

Because the hypothesized value for the population mean, $\mu_0 = 500\text{ml}$, is not in this interval, the hypothesis-testing conclusion is that the null hypothesis, $H_0\!:\ \mu = 500$, is rejected.

Thanks

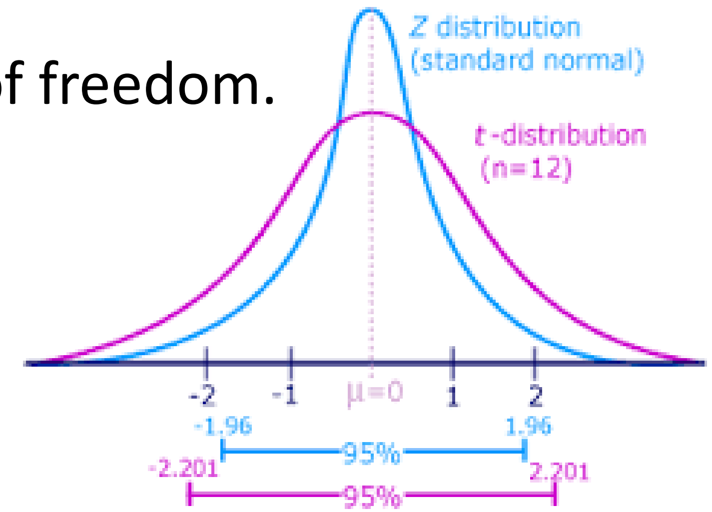
Hypothesis Testing-III

Tests About a Population Mean: σ Unknown

- Test Statistic

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

This test statistic has a *t* distribution with *n* - 1 degrees of freedom.



Tests About a Population Mean: σ Unknown

Rejection Rule: p -Value Approach

Reject H0 if p –value $\leq \alpha$

Rejection Rule: Critical Value Approach

H0: $\mu \geq \mu_0$ Reject H0 if t $\leq -t_\alpha$

H0: $\mu \leq \mu_0$ Reject H0 if t $\geq t_\alpha$

H_0: $\mu = \mu_0$ Reject H0 if t $\leq -t_{\alpha/2}$ or $t \geq t_{\alpha/2}$

In [10]:
```python
from scipy import stats
import numpy as np
```

In [11]:
```python
x=[10,12,20,21,22,24,18,15]
stats.ttest_1samp(x,15)
```

Out[11]:
```
Ttest_1sampResult(statistic=1.5623450931857947, pvalue=0.1621787560592894)
```

One-Tailed Test About a Population Mean: σ Unknown

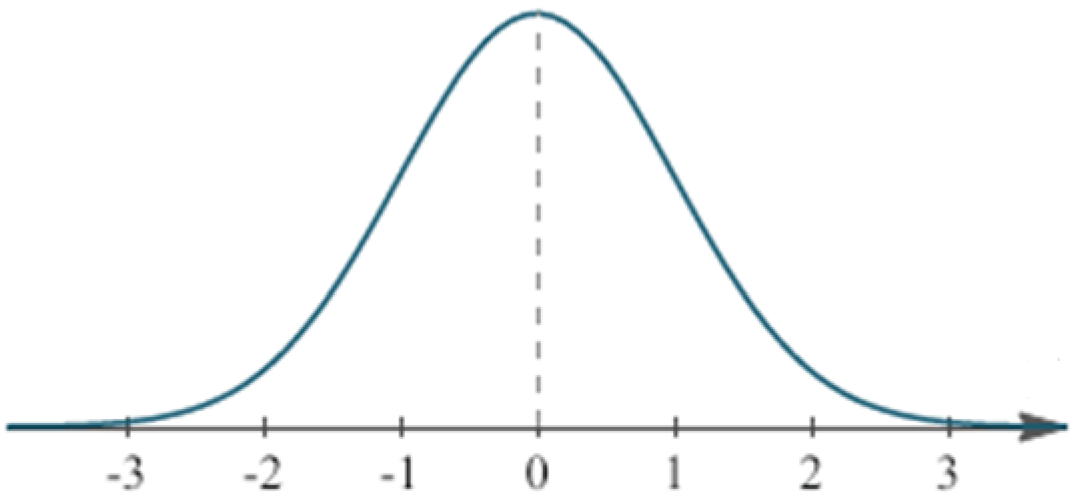
Example: Ice Cream Demand

- In a ice cream parlor at IIT Roorkee, the following data represent the number of ice-creams sold in 20 days

- Test hypothesis H_0: $\mu \le 10$
- Use α = .05 to test the hypothesis.

Day	No. of Ice-cream Sold	Day	No. of Ice-cream Sold
1	13	11	12
2	8	12	11
3	10	13	11
4	10	14	12
5	8	15	10
6	9	16	12
7	10	17	7
8	11	18	10
9	6	19	11
10	8	20	8

Given Data

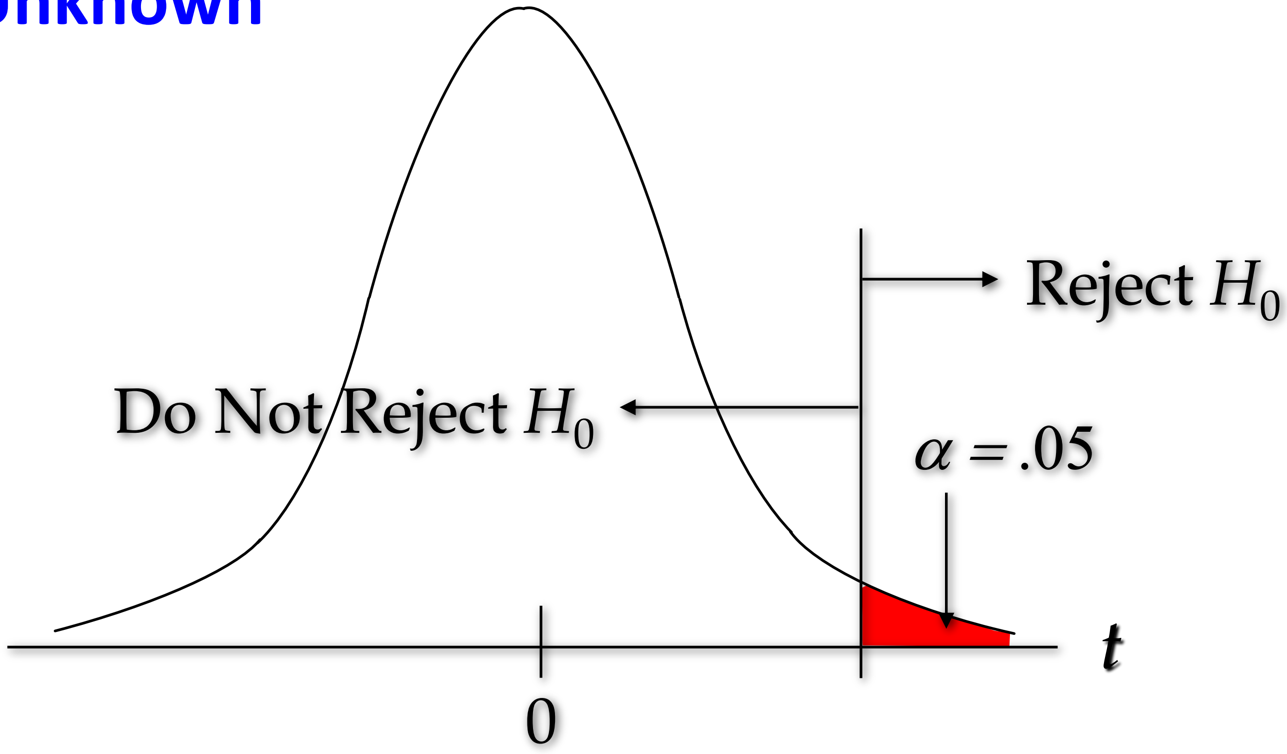


```
In [8]: x=[13,8,10,10,8,9,10,11,6,8,12,11,11,12,10,12,7,10,11,8]
```

```
In [9]: stats.ttest_1samp(x,10)
Out[9]: Ttest_1sampResult(statistic=-0.35843385854878496, pvalue=0.7239703579964252)
```

```
In [10]: 0.7239703579964252/2
Out[10]: 0.3619851789982126
```

One-Tailed Test About a Population Mean: σ Unknown



In [3]:
```
stats.t.cdf(-0.384,19)
```
Out[3]: 0.35262102566795583

[2]:
```
stats.t.ppf(0.05,19)
```
Out[2]: -1.7291328115213678

Hypothesis Testing – proportion

Null and Alternative Hypotheses: Population Proportion

- The equality part of the hypotheses always appears in the null hypothesis.
- In general, a hypothesis test about the value of a population proportion p must take one of the following three forms (where p0 is the hypothesized value of the population proportion).

One-tailed (lower tail)	One-tailed (upper tail)	Two-tailed
$H_0\colon p \geq p_0$	$H_0\colon p \leq p_0$	$H_0\colon p = p_0$
$H_a\colon p < p_0$	$H_a\colon p > p_0$	$H_a\colon p \neq p_0$

Tests About a Population Proportion

Test Statistic

$$z = \frac{\bar{p} - p_0}{\sigma_{\bar{p}}}$$

where: $$\sigma_{\bar{p}} = \sqrt{\frac{p_0(1 - p_0)}{n}}$$

assuming $np \geq 5$ and $n(1 - p) \geq 5$

Tests About a Population Proportion

Rejection Rule: *p* –Value Approach

$$\text{Reject } H_0 \text{ if } p\text{ –value} \leq \alpha$$

Rejection Rule: Critical Value Approach

$H_0\!: p \leq p_0$ Reject H_0 if $z \geq z_\alpha$

$H_0\!: p \geq p_0$ Reject H_0 if $z \leq -z_\alpha$

$H_0\!: p = p_0$ Reject H_0 if $z \leq -z_{\alpha/2}$ or $z \geq z_{\alpha/2}$

Two-Tailed Test About a Population Proportion

Example: City Traffic Police

For a New Year's week, the City Traffic Police claimed that 50% of the accidents would be caused by drunk driving.

A sample of 120 accidents showed that 67 were caused by drunk driving. Use these data to test the Traffic Police's claim with $\alpha = .05$.

p –Value Approach

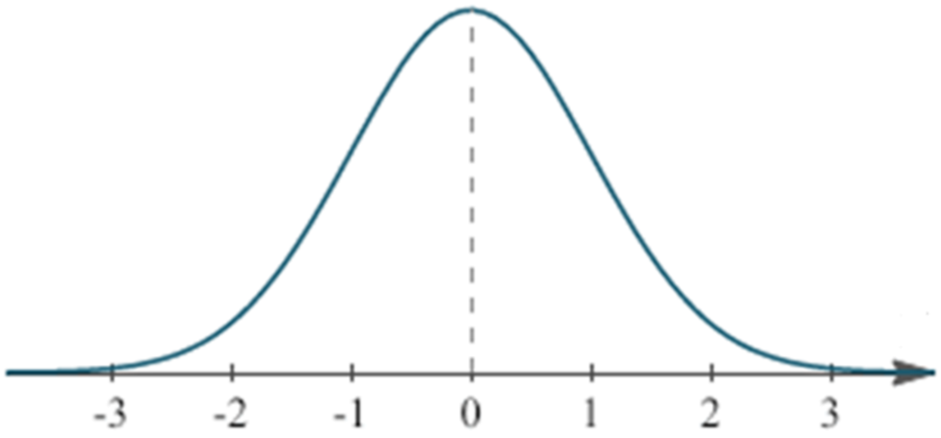
Two-Tailed Test About a Population Proportion

1. Determine the hypotheses.

$$H_0: p = .5$$

$$H_a: p \neq .5$$

2. Specify the level of significance. $\alpha = .05$



3. Compute the value of the test statistic.

$$\sigma_{\bar{p}} = \sqrt{\frac{p_0(1-p_0)}{n}} = \sqrt{\frac{.5(1-.5)}{120}} = .045644$$

$$z = \frac{\bar{p} - p_0}{\sigma_{\bar{p}}} = \frac{(67/120) - .5}{.045644} = 1.28$$

Two-Tailed Test About a Population Proportion

4. Compute the p -value.

For $z = 1.28$, cumulative probability = .8997 p–value = 2(1 - .8997) = .2006

5. Determine whether to reject H0.

Because p–value = .2006 > α = .05, we cannot reject H0.

```
In [13]: from statsmodels.stats.proportion import proportions_ztest

In [14]: count=67

In [16]: samplesize = 120

In [17]: P=0.5

In [18]: proportions_ztest(count, samplesize,P)

Out[18]: (1.286806739751111, 0.19816165722384 55)
```

Critical Value Approach

Two-Tailed Test About a Population Proportion

4. Determine the critical value and rejection rule.

For $\alpha/2 = .05/2 = .025$, $z_{.025} = 1.96$

Reject H_0 if $z \leq -1.96$ or $z \geq 1.96$

5. Determine whether to reject H0.

Because 1.278 > -1.96 and < 1.96, we cannot reject H0.

Errors in Hypothesis Testing

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Example

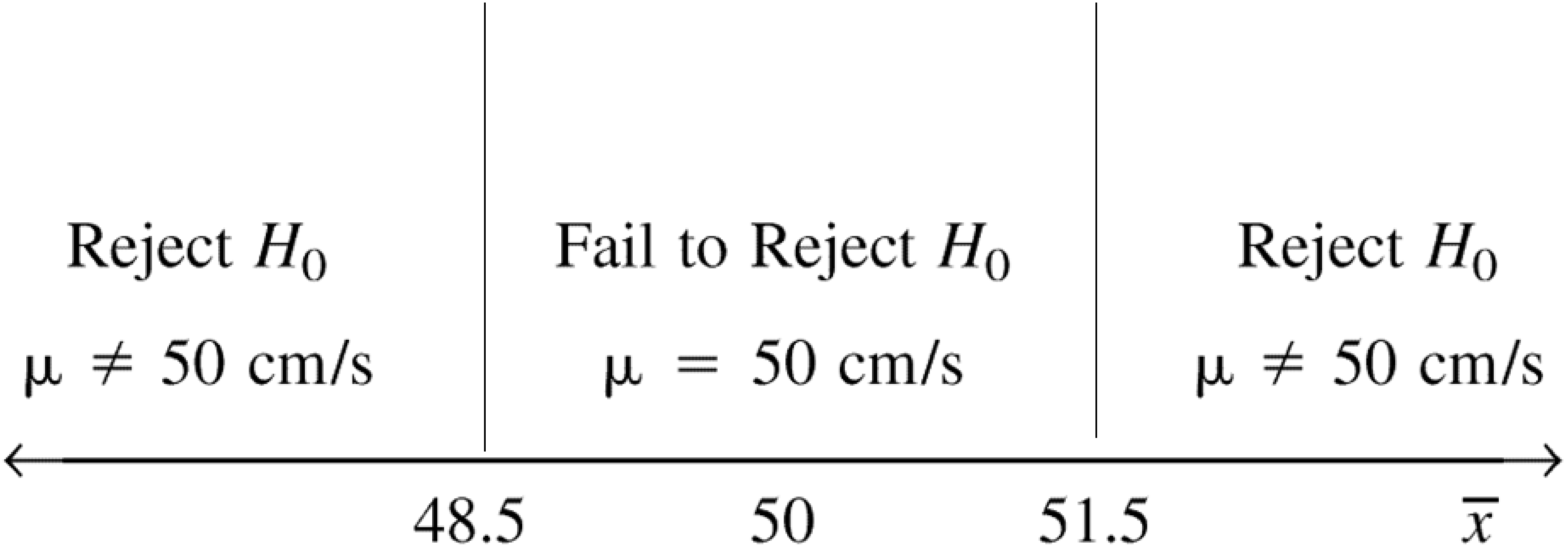
- We are interested in burning rate of a solid propellant used to power aircrew escape systems
- Burning rate is a random variable that can be described by a probability distribution
- Suppose our interest focus on mean burning rate
- Ho: $\mu = 50$ centimeters per second
- H1: $\mu \neq 50$ centimeters per second

Reference: Applied statistics and probability for engineers, Douglas C. Montgomery, George C. Runger, John Wiley & Sons, 2007

Value of the null hypothesis

- The value of the null hypothesis can be obtained by
  - Past experience or knowledge of the process, or even from the previous tests or experiments
  - From some theory or model regarding the process under study
  - From external consideration, such as design or engineering specifications, or from contractual obligations

Note: for this example n=10



Decision criteria for testing H_0: $\mu = 50$ cm/s versus H_1: $\mu \neq 50$ cm/s.

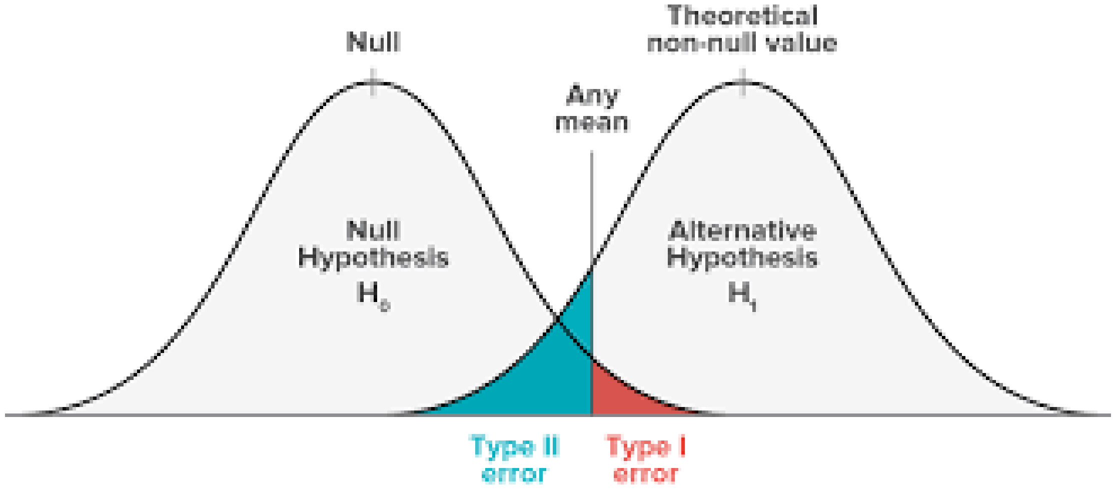
Note: for this example we will assume $\sigma = 2.5$

Type I Error

- The true mean burning rate of the propellant could be equal to 50 centimeters per second
- However randomly selected propellant specimens that are tested, we could observe a value of test statistics $\bar{x}$ that falls into the critical region(rejection region).
- We would then reject the null hypothesis Ho in favor of the alternate H1, in fact, Ho is really true
- This type of wrong conclusion is called a type I error

Type I Error

- Rejecting the null hypothesis Ho when it is true is defined as a type I error

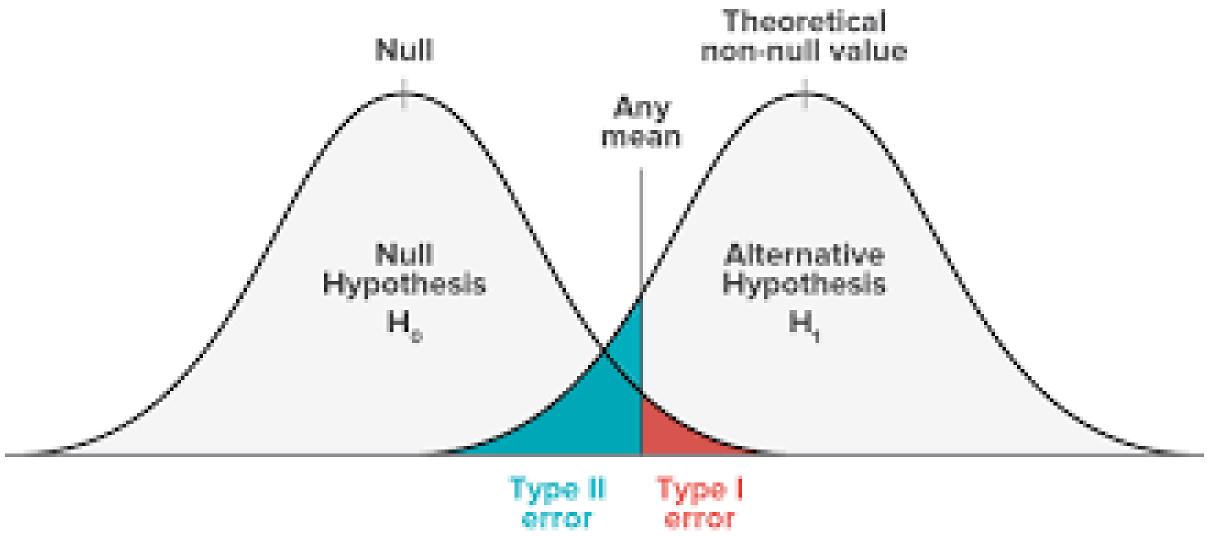


Type II Error

- Now suppose the true mean burning rate is different from 50 centimeters per second, yet the sample mean $\bar{x}$ falls in the acceptance region
- In this case we would fail to reject Ho when it is false
- This type of wrong conclusion is called a type II error

Type II Error

- Failing to reject the null hypothesis when it is false is defined as a type II error



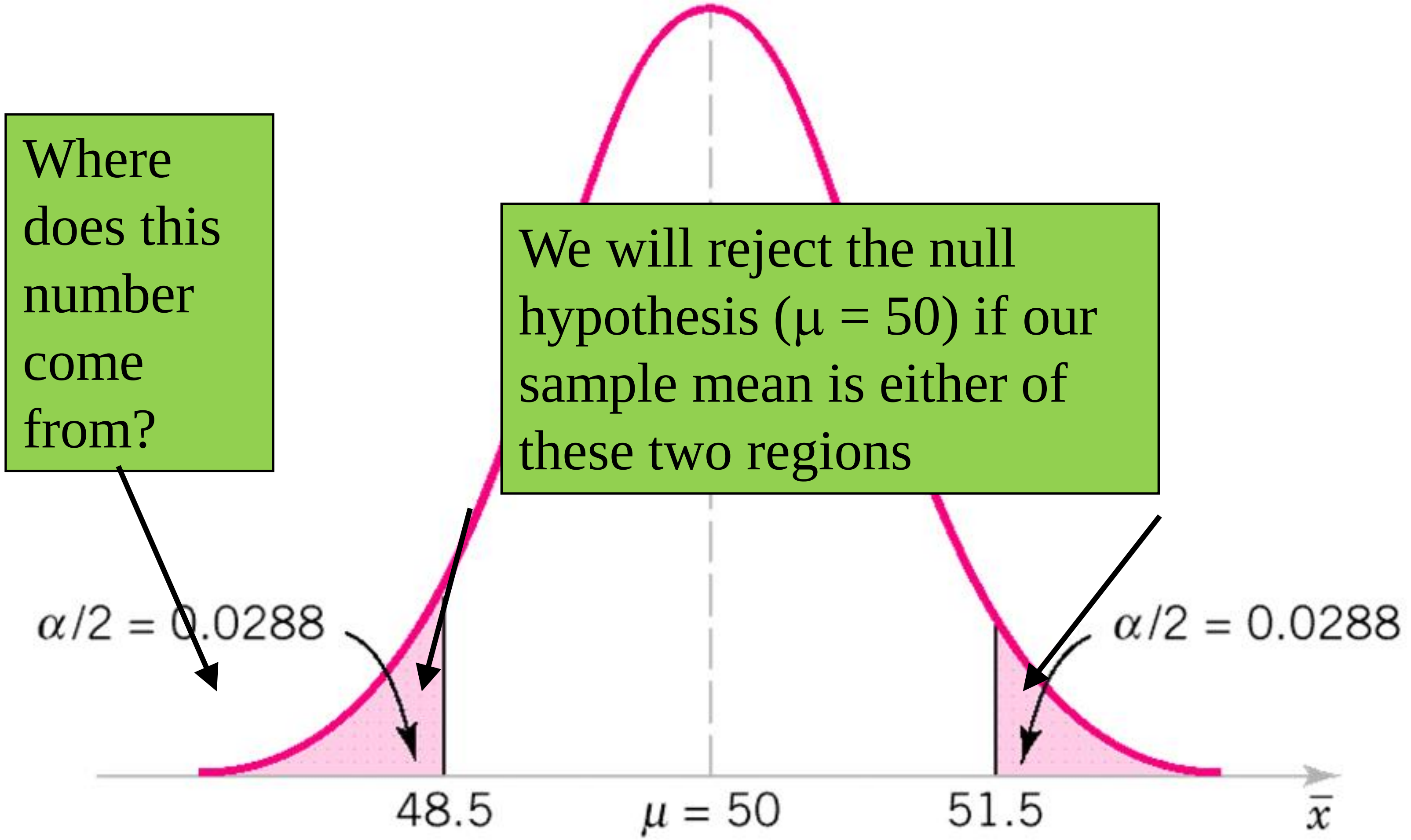
Type 1 and Type II Errors

	H_0 is correct	H_0 is incorrect
H_0 is accepted	correct decision	Type II error (β) **Incorrect acceptance**
H_0 is rejected	Type I error (α) **Incorrect rejection**	correct decision

Type I error

- In the propellant burning rate example, a type I error will occur when either $\bar{x} > 51.5\_or\_\bar{x} < 48.5$ when the true mean burning rate is μ = 50 centimeters per second
- Suppose the standard deviation of burning rate is σ = 2.5 centimeters per second and n = 10
- Probability distribution μ = 50,standard error = 0.79.
- Type I error is

$$\alpha = P(\bar{x} < 48.5\_when\_\mu = 50) + P(\bar{x} > 51.5\_when\_\mu = 50)$$



The critical region for H_0: $\mu = 50$ versus H_1: $\mu \neq 50$ and $n = 10$.

Defing function for calculating alpha value

In [6]:
```
def z_value(x,mu,SEM):
 z = (x - mu)/SEM
 if(z < 0):
  alfa = stats.norm.cdf(z)
 else:
  alfa = 1 - stats.norm.cdf(z)
 print (alfa)
```

calculating aplha for different values of x,mu, and SEM

In [8]:
```
x =48.5
mu = 50
SEM = 0.79
```

In [9]:
```
z_value(x,mu,SEM)
```

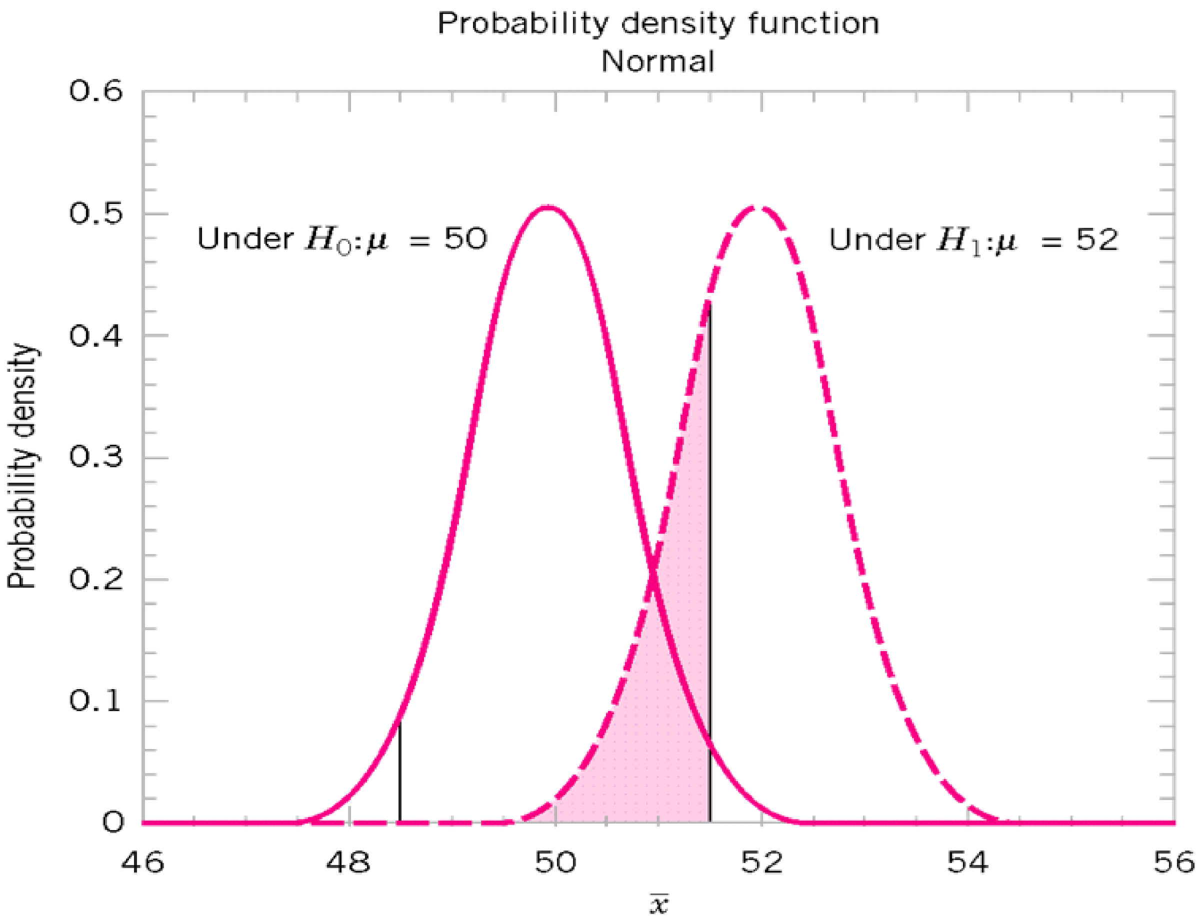
```
0.02879971774715278
```

Type I error

- Type I error = 0.057434
- This implies that 5.7 % of all random samples would lead to rejection of the hypothesis Ho: μ=50 centimeters per second.
- We can reduce the type I error by widening the acceptance region. If we make critical value 48 and 52, the value of alpha is 0.0114 (adding 0.0057 and 0.0057).
- Change sample size to 16 then alpha is 0.0164.

```
In [40]: z_value(48,mu,SEM)

0.005676434117424844

In [41]: z_value(52,mu,SEM)

0.0056764341174248
```

TYPE II ERROR



The probability of type II error when $\mu = 52$ and $n = 10$.

The pink area is the probability

of a Type II error if the actual mean is 52.

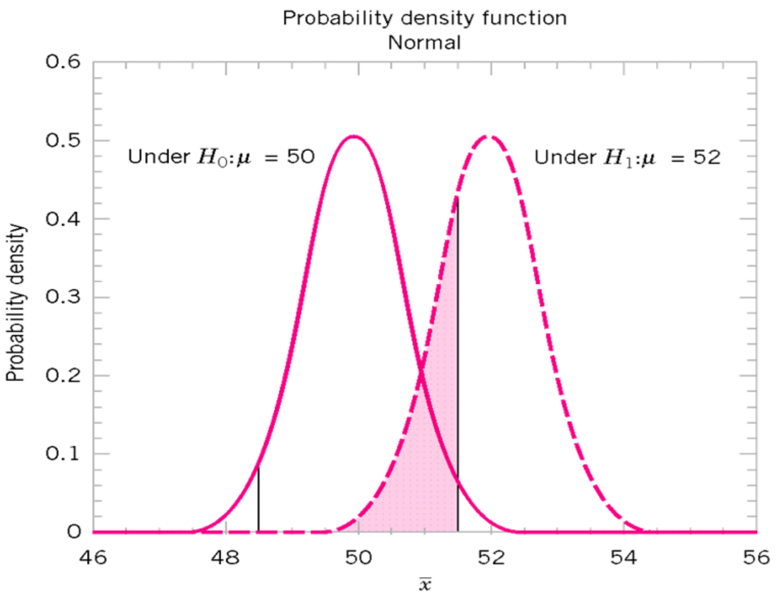
Type II Error

- Type II error will be committed if the sample mean x-bar falls between 48.5 and 51.5 (critical region boundaries) when μ = 52. $\beta = P(48.5 \leq \bar{x} \leq 51.5\_when\_\mu = 52)$

- 0.2643

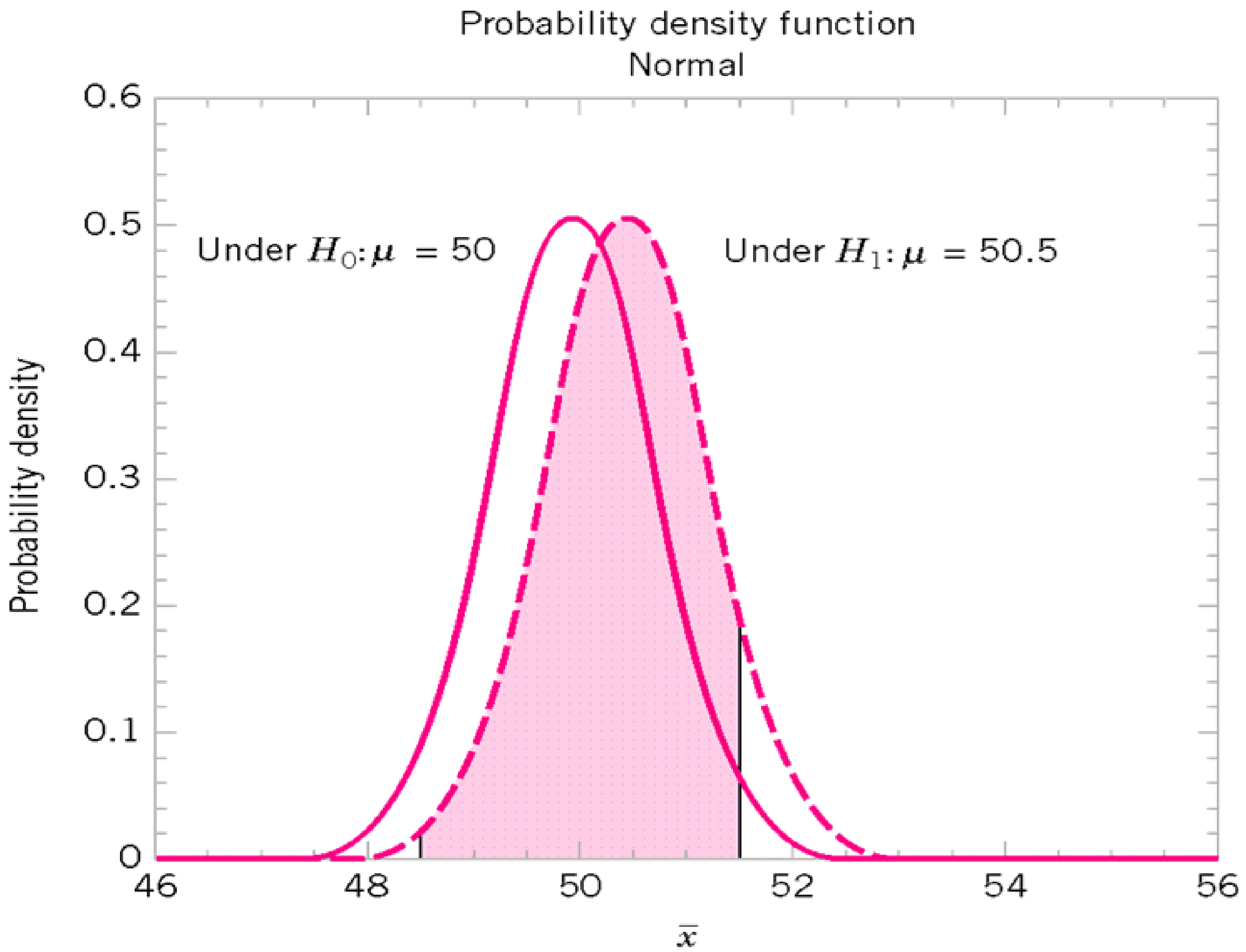
- When μ = 50.5

- 0.8923

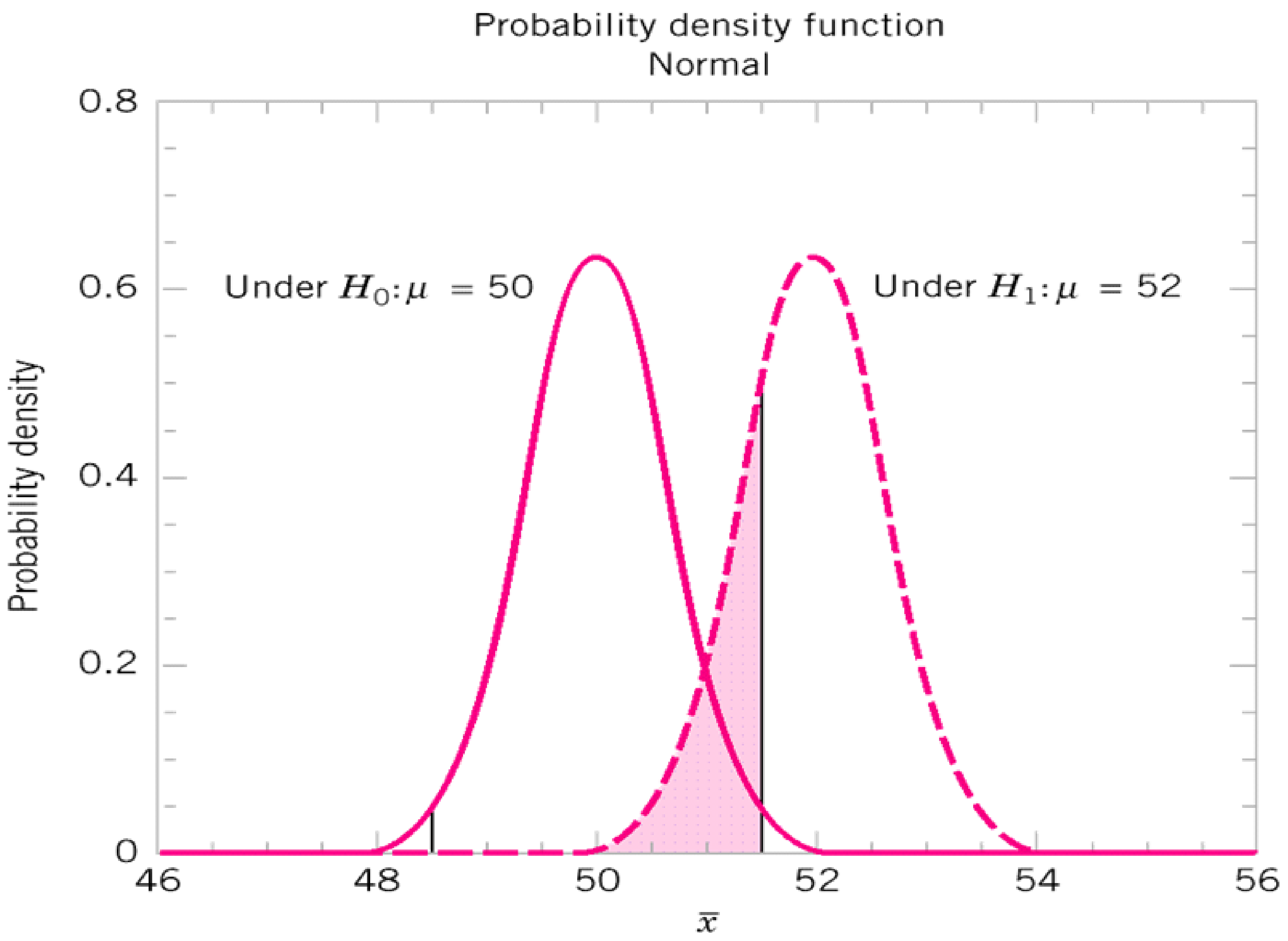


The probability of type II error when μ = 52 and n = 10.

```
In [4]: beta = stats.norm.cdf((51.5-52)/0.79) #

In [5]: beta
Out[5]: 0.26339575390741593

In [8]: beta = stats.norm.cdf((51.5-50.5)/0.79) #

In [9]: beta
Out[9]: 0.8972117321157791
```



The probability of type II error when $\mu = 50.5$ and $n = 10$.



The probability of type II error when $\mu = 52$ and $n = 16$.

Computing the probability of a type II error may be the most difficult concept

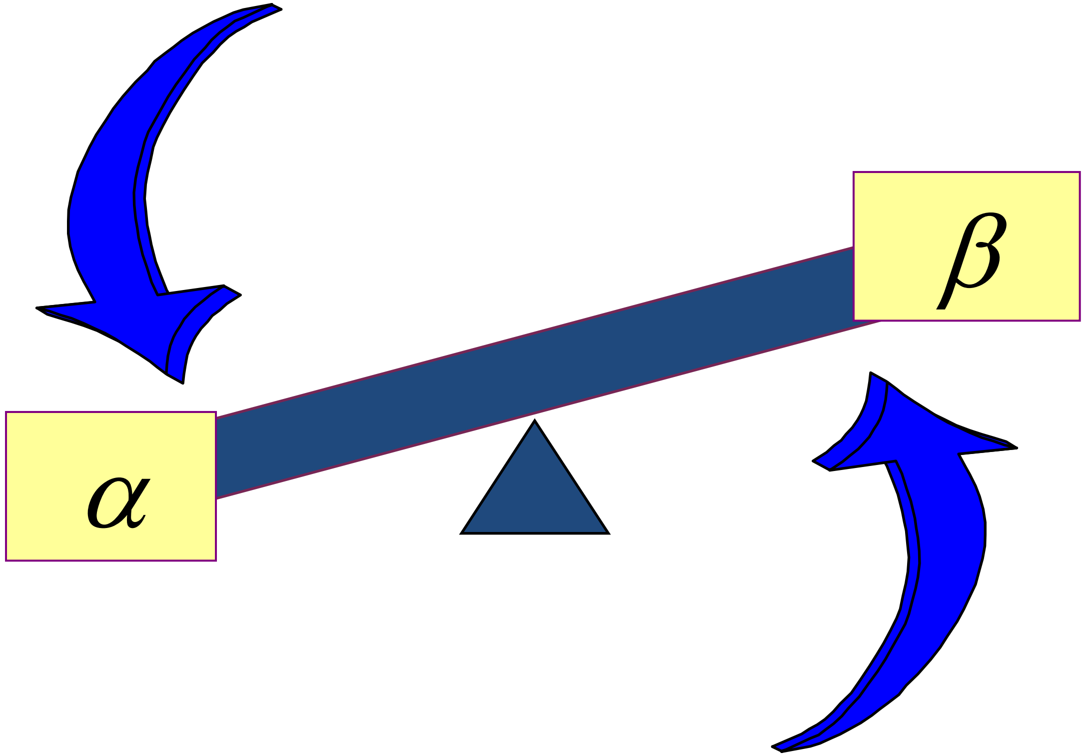
acceptance region	sample size	α	β at $\mu = 52$	β at $\mu = 50.5$
$48.5 < \bar{x} < 51.5$	10	0.0576	0.2643	0.8923
$48 < \bar{x} < 52$	10	0.0114	$\boxed{0.5000}$	$\boxed{0.9705}$
$48.5 < \bar{x} < 51.5$	16	0.0164	0.2119	$\boxed{0.9445}$
$48 < \bar{x} < 52$	16	$\boxed{0.0014}$	$\boxed{0.5000}$	$\boxed{0.9918}$

For constant n, increasing the acceptance region (hence decreasing α) increases β.

Increasing n, can decrease both types of errors.

Type I & II Errors Have an Inverse Relationship

If you reduce the probability of one error, the other one increases so that everything else is unchanged.



Factors Affecting Type II Error

- True value of population parameter
  - β Increases when the difference between hypothesized parameter and its true value decrease
- Significance level α
  - Increases when β decreases
- Population standard deviation σ
  - Increases when β increases
- Sample size
  - β Increases when n decreases

$\beta \uparrow \quad \alpha \downarrow$

$\beta \uparrow \quad \sigma \uparrow$

$\beta \uparrow \quad n \downarrow$

How to Choose between Type I and Type II Errors

- Choice depends on the cost of the errors
- Choose smaller Type I Error when the cost of rejecting the maintained hypothesis is high
  – A criminal trial: convicting an innocent person
- Choose larger Type I Error when you have an interest in changing the status quo

Calculating the probability of Type II Error

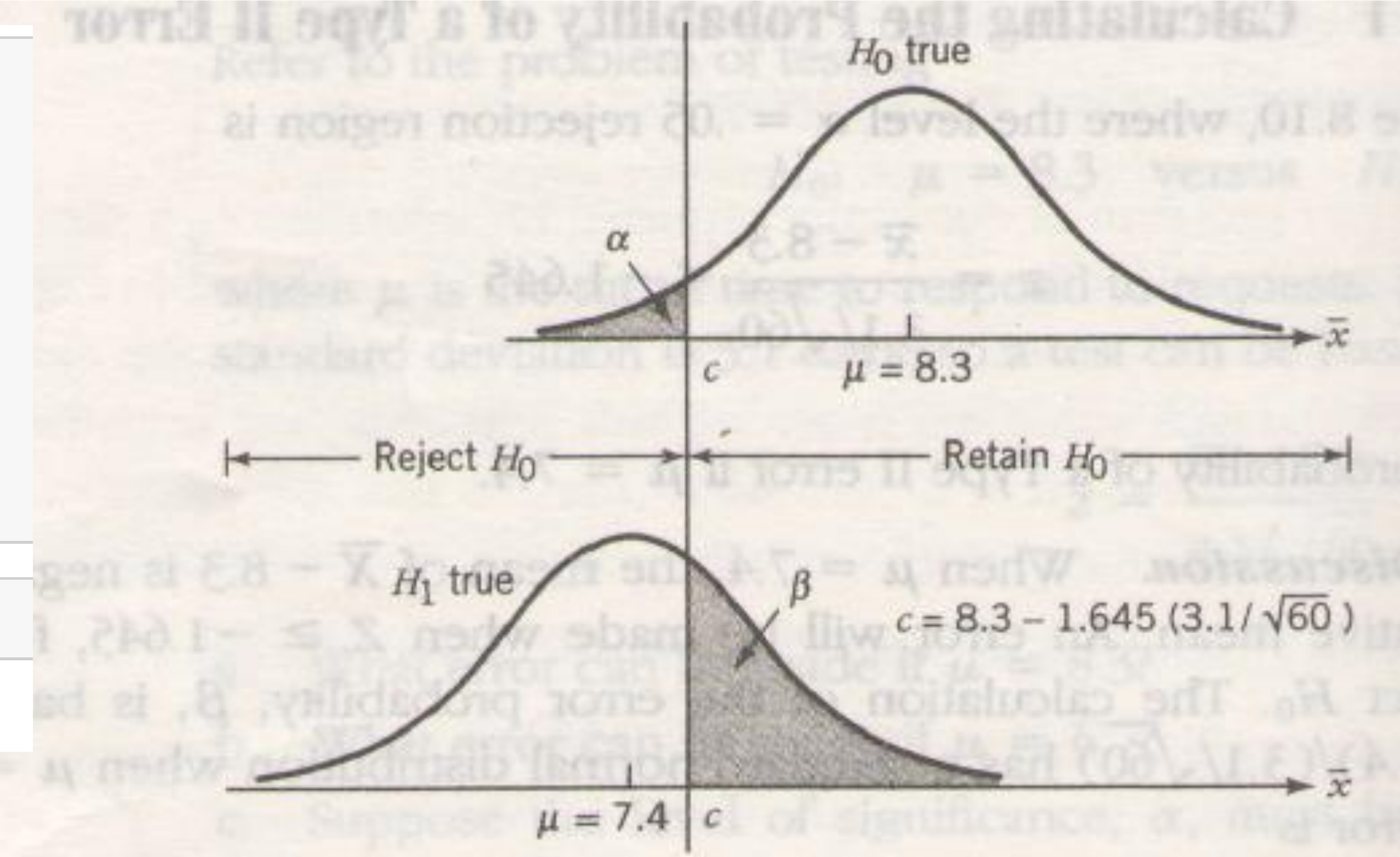
Ho: $\mu = 8.3$
H1: $\mu < 8.3$

Determine the probability of Type II error if $\mu = 7.4$ at 5% significance level. $\sigma = 3.1$ and $n = 60$.

Solution:

```
In [48]: def type_2(mu1,mu2,sigma,n,alfa):
          z =stats.norm.ppf(alfa)
          xbar = (mu1)+(z*sigma/np.sqrt(n))
          z2 = (xbar - mu2)/(sigma/np.sqrt(n))
          if(mu1 > mu2):
           beta= 1-stats.norm.cdf(z2)
          else:
           beta = stats.norm.cdf(z2)

          print (beta)
```

```
In [49]: type_2(8.3,7.4,3.1,60,0.05)

0.27292999450730004
```



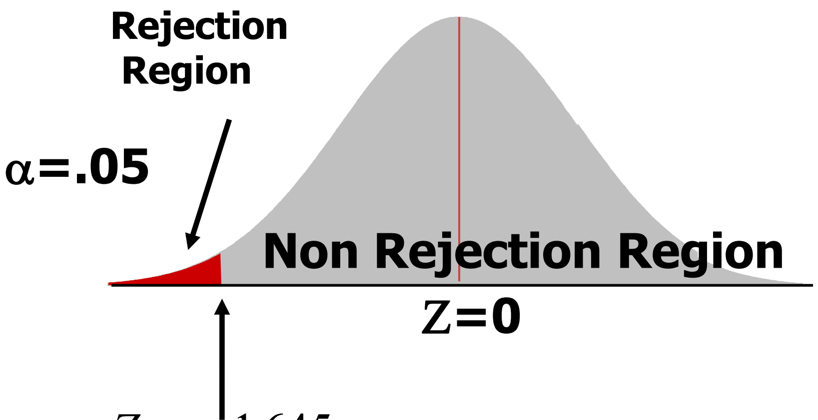
An error will be made when Z ≥ -1.645, for that will fail to reject Ho.

β = 0.2729

Solving for Type II Errors: Example

$$H_o: \mu = 12$$

$$H_a: \mu < 12$$



$$Z_c = -1.645$$

$$\overline{X}_c = \mu + Z_c \frac{\sigma}{\sqrt{n}}$$

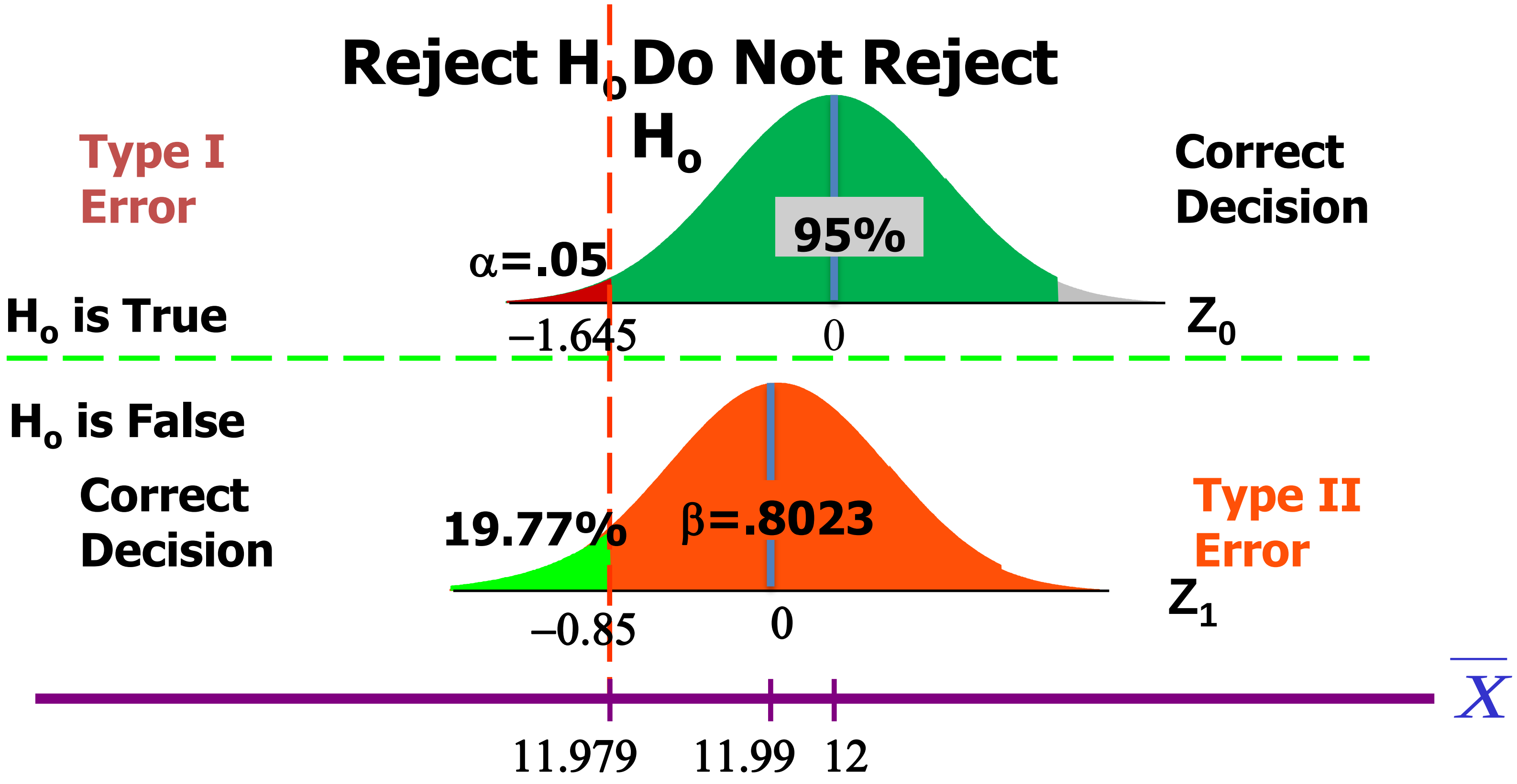
$$= 12 + (-1.645)\frac{0.10}{\sqrt{60}}$$

$$= 11.979$$

If $\overline{X} < 11.979$, reject H_o.

If $\overline{X} \geq 11.979$, do not reject H_o.

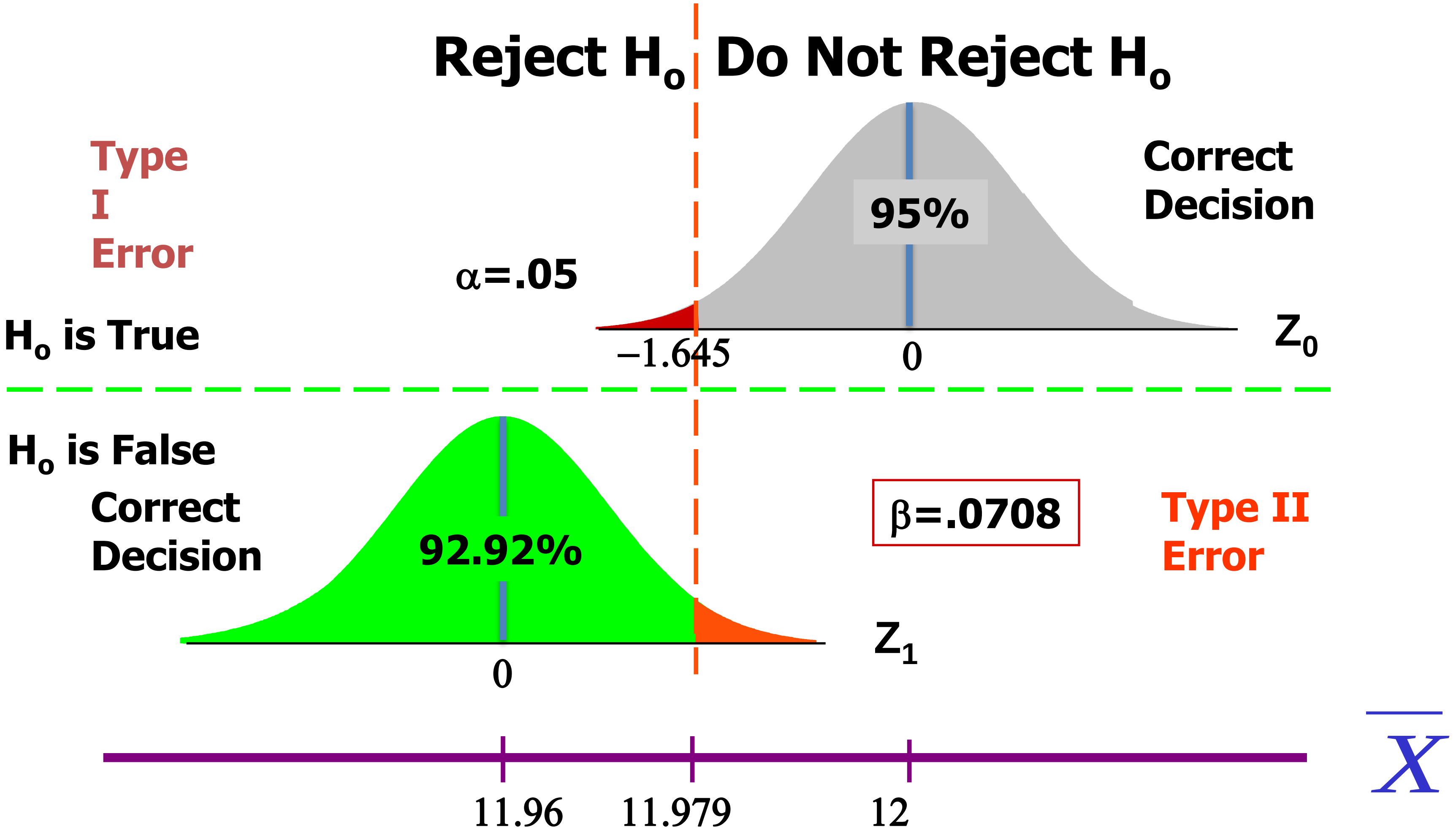
Type II Error for Example with μ =11.99 Kg



In [50]:
```
type_2(12,11.99,0.1,60,0.05)
```

```
0.8079200023112734
```

Type II Error for Demonstration with μ=11.96 Kg



```
In [51]: type_2(12,11.96,0.1,60,0.05)
```

```
0.07303790512847008
```

Hypothesis Testing and Decision Making

- We have illustrated hypothesis testing applications referred to as significance tests

- In the tests, we compared the *p*-value to a controlled probability of a Type I error, a, which is called the level of significance for the test

- With a significance test, we control the probability of making the Type I error, but not the Type II error

- We recommended the conclusion "do not reject H_0" rather than "accept H_0" because the latter puts us at risk of making a Type II error

Hypothesis Testing and Decision Making

- With the conclusion "do not reject H_0", the statistical evidence is considered inconclusive
- Usually this is an indication to postpone a decision until further research and testing is undertaken
- In many decision-making situations the decision maker may want, and in some cases may be forced, to take action with both the conclusion "do not reject H_0 "and the conclusion "reject H_0."
- In such situations, it is recommended that the hypothesis-testing procedure be extended to include consideration of making a Type II error

Power of a test

- The mean response time for a random sample of 40 food-order is 13.25 minutes
- The population standard deviation is believed to be 3.2 minutes.
- The restaurant owner wants to perform a hypothesis test, with α =0.05 level of significance, to determine whether the service goal of 12 minutes or less is being achieved.

Calculating the Probability of a Type II Error

Hypotheses are: H_0: $\mu \leq 12$ and H_a: $\mu > 12$

Rejection rule is: Reject H_0 if $z \geq 1.645$

Value of the sample mean that identifies the rejection region:

$$z = \frac{\bar{x} - 12}{3.2/\sqrt{40}} \geq 1.645$$

$$\bar{x} \geq 12 + 1.645\left(\frac{3.2}{\sqrt{40}}\right) = 12.8323$$

We will accept H_0 when $x \leq 12.8323$

Calculating the Probability of a Type II Error

Probabilities that the sample mean will be in the acceptance region:

Values of μ	$z = \dfrac{12.8323 - \mu}{3.2/\sqrt{40}}$	β	$1\text{-}\beta$
14.0	-2.31	.0104	.9896
13.6	-1.52	.0643	.9357
13.2	-0.73	.2327	.7673
12.8323	0.00	.5000	.5000
12.8	0.06	.5239	.4761
12.4	0.85	.8023	.1977
12.0001	1.645	.9500	.0500

```
In [20]: type_2(14,12,3.2,40,0.05)
0.010499750448532241

In [21]: type_2(13.6,12,3.2,40,0.05)
0.06457982995225997

In [23]: type_2(13.2,12,3.2,40,0.05)
0.2336575101104159

In [22]: type_2(12.8323,12,3.2,40,0.05)
0.49995065746353273

In [27]: type_2(12.8,12,3.2,40,0.05)
0.5254013387545549

In [24]: type_2(12.4,12,3.2,40,0.05)
0.8035262335707292

In [26]: type_2(12.0001,12,3.2,40,0.05)
0.9499796127157129
```

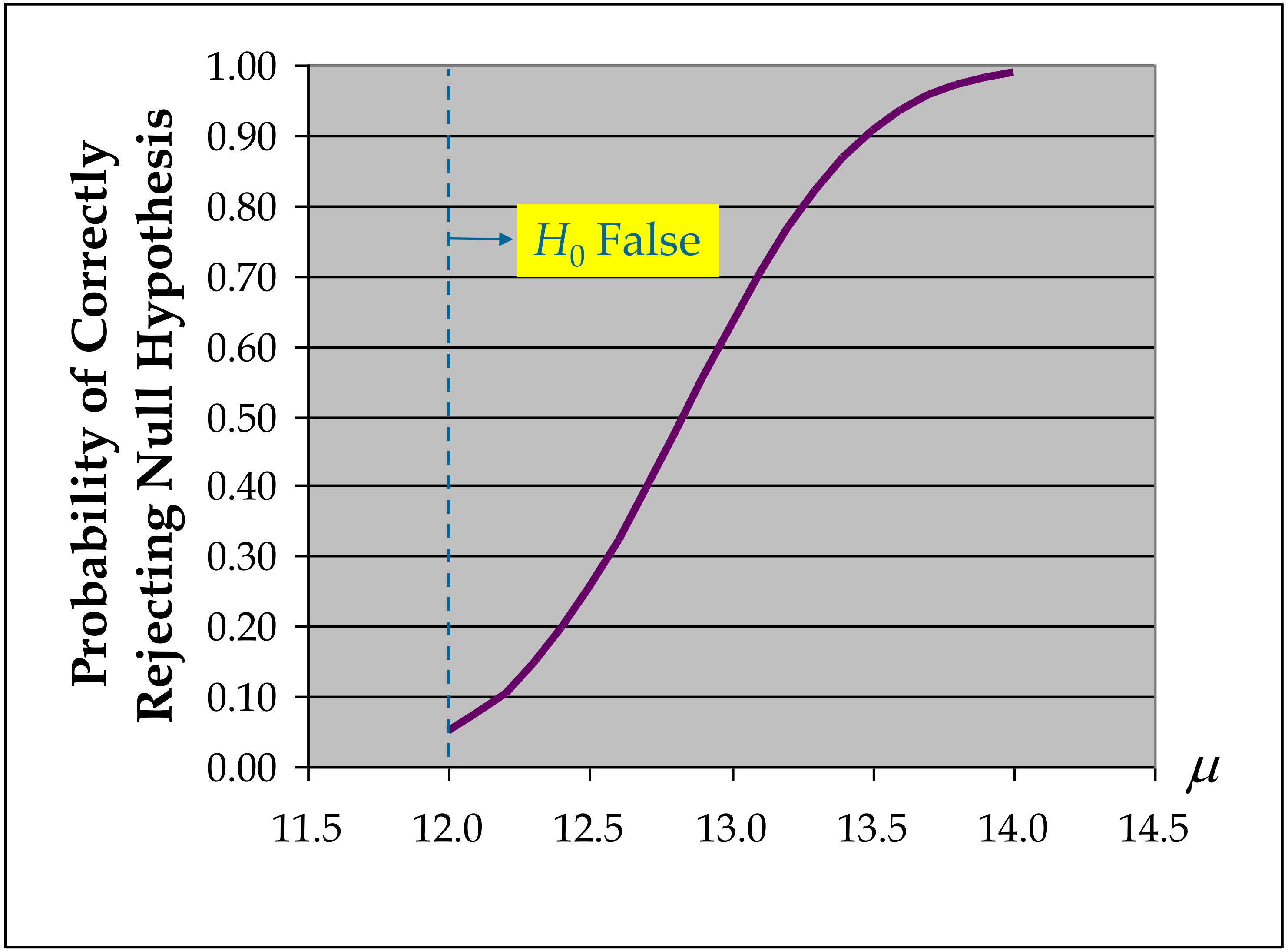
Power of the Test

- The probability of correctly rejecting H_0 when it is false is called the power of the test.

- For any particular value of m, the power is $1 - b$.

- We can show graphically the power associated with each value of μ ; such a graph is called a power curve.

Power Curve


1.00
0.90
0.80
0.70
0.60
0.50
0.40
0.30
0.20
0.10
0.00
Probability of Correctly Rejecting Null Hypothesis
H_0 False
11.5
12.0
12.5
13.0
13.5
14.0
14.5
μ

Thank You

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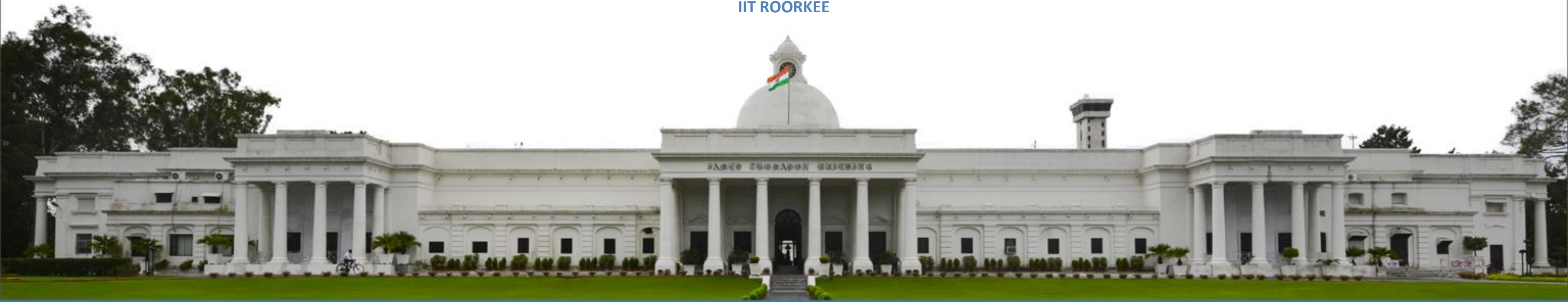
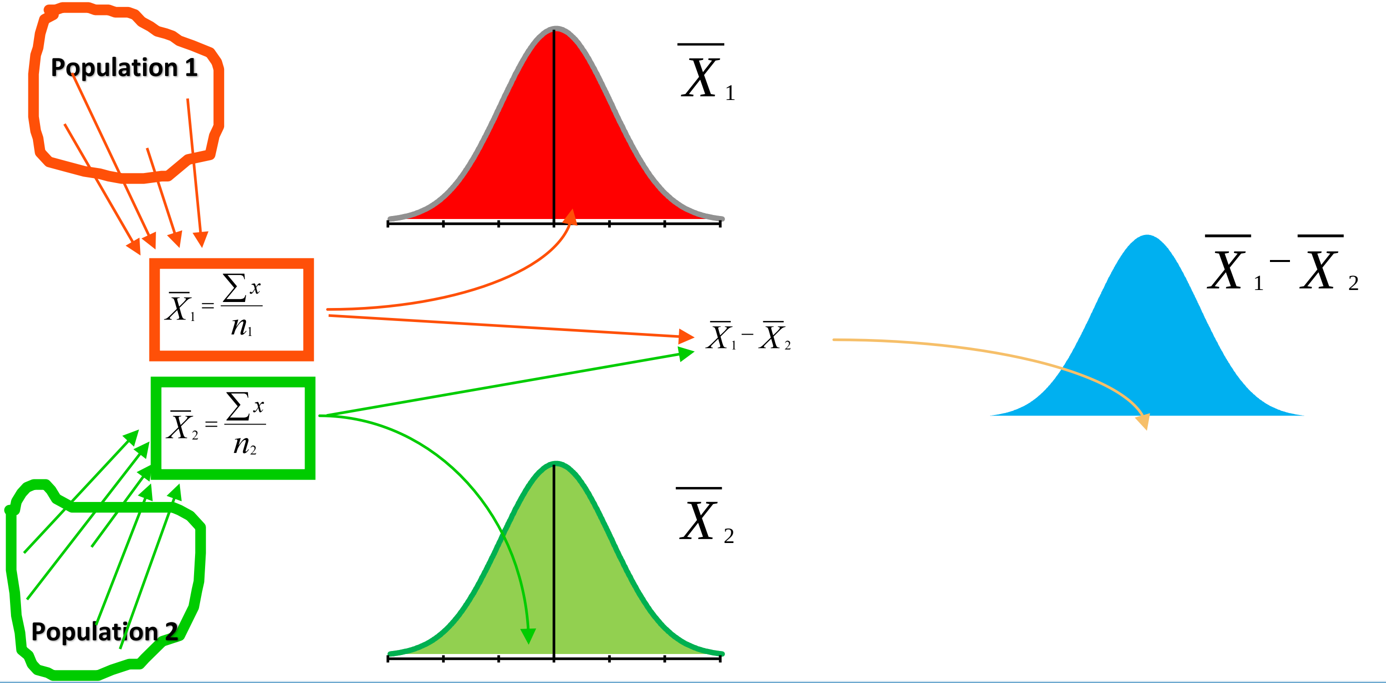
NPTEL ONLINE CERTIFICATION COURSE

Hypothesis Testing: Two sample test

Dr. A. Ramesh
DEPARTMENT OF MANAGEMENT
IIT ROORKEE



Hypothesis Testing about the Difference in Two Sample Means



Two Sample Tests

Two Sample Tests

- Population Means, Independent Samples
- Population Means, Dependent Samples
- Population Proportions
- Population Variances

Examples:

- Group 1 vs. independent Group 2
- Same group before vs. after treatment
- Proportion 1 vs. Proportion 2
- Variance 1 vs. Variance 2

Difference Between Two Means

- Population means, independent samples
  - σ_1^2 and σ_2^2 known → Test statistic is a z value
  - σ_1^2 and σ_2^2 unknown
    - σ_1^2 and σ_2^2 assumed equal → Test statistic is a value from the Student's t distribution
    - σ_1^2 and σ_1^2 assumed unequal → Test statistic is a value from the Student's t distribution

σ_1^2 and σ_1^2 Known

Population means, independent samples

σ_1^2 and σ_2^2 known

σ_1^2 and σ_2^2 unknown

Assumptions:

- Samples are randomly and independently drawn
- both population distributions are normal
- Population variances are known

σ_1^2 and σ_2^2 Known

Population means, independent samples

- σ_1^2 and σ_2^2 known
- σ_1^2 and σ_2^2 unknown

When σ_x^2 and σ_y^2 are known and both populations are normal, the variance of 1 – 2 is

$$\boxed{\sigma^2_{\overline{X}_1-\overline{X}_2} = \frac{\sigma_1^{\ 2}}{n_1} + \frac{\sigma_2^{\ 2}}{n_2}}$$

…and the random variable

$$\boxed{Z = \frac{(\overline{x}_1 - \overline{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}}$$

has a standard normal distribution

Test Statistic, σ_1^2 and σ_2^2 Known

Population means, independent samples

- σ_1^2 and σ_2^2 known
- σ_1^2 and σ_2^2 unknown

$$H_0: \mu_1 - \mu_2 = D_0$$

The test statistic for $\mu_1 - \mu_2$ is:

$$z = \frac{\left(\bar{x}_1 - \bar{x}_2\right) - D_0}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Hypothesis Tests for Two Population Means

Two Population Means, Independent Samples

Lower-tail test:

$H_0: \mu_1 \geq \mu_2$

$H_1: \mu_1 < \mu_2$

i.e.,

$H_0: \mu_1 - \mu_2 \geq 0$

$H_1: \mu_1 - \mu_2 < 0$

Upper-tail test:

$H_0: \mu_1 \leq \mu_2$

$H_1: \mu_1 > \mu_2$

i.e.,

$H_0: \mu_1 - \mu_2 \leq 0$

$H_1: \mu_1 - \mu_2 > 0$

Two-tail test:

$H_0: \mu_1 = \mu_2$

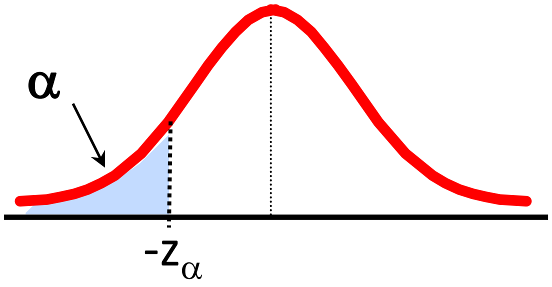
$H_1: \mu_1 \neq \mu_2$

i.e.,

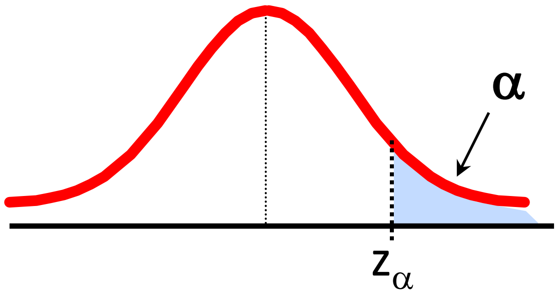
$H_0: \mu_1 - \mu_2 = 0$

$H_1: \mu_1 - \mu_2 \neq 0$

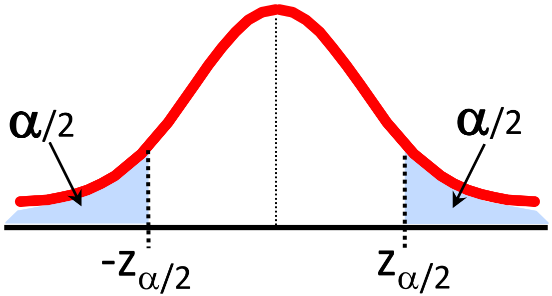
Decision Rules



Reject H_0 if $z < -z_{\alpha}$



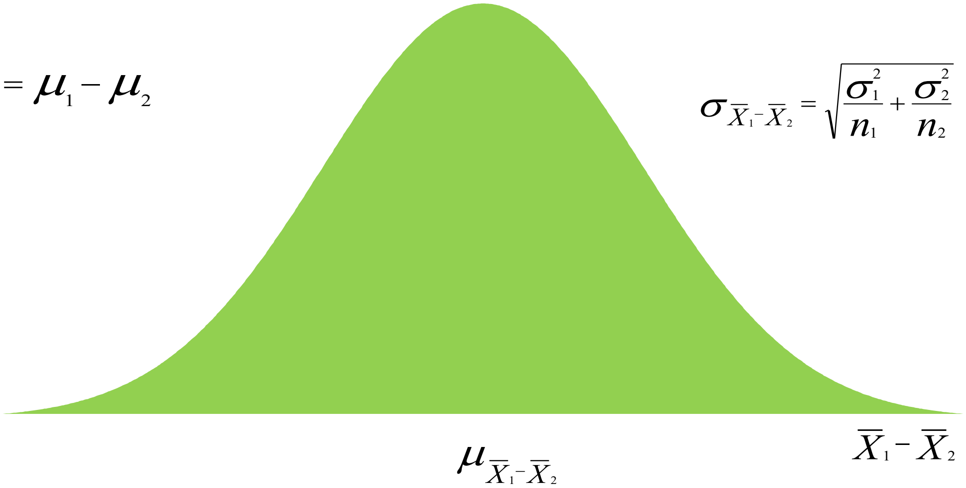
Reject H_0 if $z > z_{\alpha}$



Reject H_0 if $z < -z_{\alpha/2}$ or $z > z_{\alpha/2}$

Hypothesis Testing about the Difference in Two Sample Means

$$\mu_{\bar{X}_1-\bar{X}_2} = \mu_1 - \mu_2$$



$$\sigma_{\bar{X}_1-\bar{X}_2} = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

Sampling Distribution of $\bar{x}_1 - \bar{x}_2$

- Expected Value $$E(\bar{x}_1 - \bar{x}_2) = \mu_1 - \mu_2$$

- Standard Deviation (Standard Error) $$\sigma_{\bar{x}_1 - \bar{x}_2} = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

where: σ_1 = standard deviation of population 1
σ_2 = standard deviation of population 2
n_1 = sample size from population 1
n_2 = sample size from population 2

Interval Estimation of μ_1 - μ_2: σ_1 and σ_2 Known

- Interval Estimate

$$\bar{x}_1 - \bar{x}_2 \pm z_{\alpha/2}\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

where: $1 - \alpha$ is the confidence coefficient

Problem (σ_1 and σ_2 Known)

- A product developer is interested in reducing the drying time of a primer paint.
- Two formulations of the paint are tested; formulation 1 is the standard chemistry, and formulation 2 has a new drying ingredient that should reduce the drying time.
- From experience, it is known that the standard deviation of drying time is 8 minutes, and this inherent variability should be unaffected by the addition of the new ingredient.
- Ten specimens are painted with formulation 1, and another 10 specimens are painted with formulation 2; the 20 specimens are painted in random order.
- The two-sample average drying times are $\overline{x_1} = 121$ minutes and $\overline{x_2}$ = 112 minutes, respectively.
- What conclusions can the product developer draw about the effectiveness of the new ingredient, using alpha = 0.05?

Source: Applied Probability and statistics for Engineers by Douglas C. Montgomery and George C. Runger *John Wiley, 3rd Ed. 2003*

Problem (σ_1 and σ_2 Known)

1. The quantity of interest is the difference in mean drying times, $\mu_1 - \mu_2$, and $\Delta_0 = 0$.
2. H_0: $\mu_1 - \mu_2 = 0$, or H_0: $\mu_1 = \mu_2$.
3. H_1: $\mu_1 > \mu_2$. We want to reject H_0 if the new ingredient reduces mean drying time.
4. $\alpha = 0.05$
5. The test statistic is

$$z_0 = \frac{\bar{x}_1 - \bar{x}_2 - 0}{\sqrt{\dfrac{\sigma_1^2}{n_1} + \dfrac{\sigma_2^2}{n_2}}}$$

where $\sigma_1^2 = \sigma_2^2 = (8)^2 = 64$ and $n_1 = n_2 = 10$.

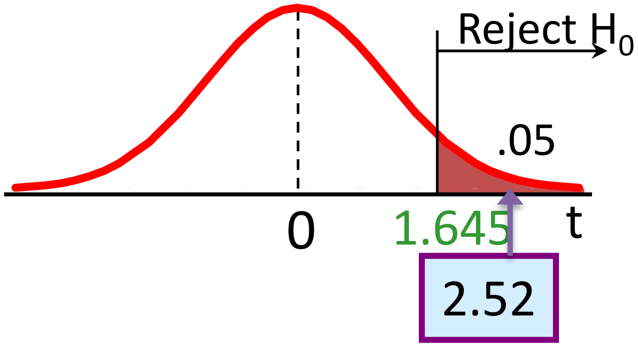
Problem (σ_1 and σ_2 Known)

6. Reject H_0: $\mu_1 = \mu_2$ if $z_0 > 1.645 = z_{0.05}$.
7. Computations: Since $\bar{x}_1 = 121$ minutes and $\bar{x}_2 = 112$ minutes, the test statistic is

$$z_0 = \frac{121 - 112}{\sqrt{\frac{(8)^2}{10} + \frac{(8)^2}{10}}} = 2.52$$

Problem (σ_1 and σ_2 Known)

$$t = \frac{(121 - 112) - 0}{\sqrt{8^2\left(\frac{1}{10} + \frac{1}{10}\right)}} = 2.52$$



Decision:

Reject H_0 at $\alpha = 0.05$

Conclusion:

There is evidence of a difference in means.

Problem (σ_1 and σ_2 Known)

8. Conclusion: Since $z_0 = 2.52 > 1.645$, we reject H_0: $\mu_1 = \mu_2$ at the $\alpha = 0.05$ level and conclude that adding the new ingredient to the paint significantly reduces the drying time. Alternatively, we can find the P-value for this test as

$$P\text{-value} = 1 - \Phi(2.52) = 0.0059$$

Therefore, H_0: $\mu_1 = \mu_2$ would be rejected at any significance level $\alpha \geq 0.0059$.

Problem (σ_1 and σ_2 Known)

In [2]:
```
import pandas as pd
import numpy as np
import math
from scipy import stats
```

In [6]:
```
def Z_and_p(x1,x2,sigma1,sigma2,n1,n2):
    z = (x1-x2)/(math.sqrt(((sigma1**2)/n1)+((sigma2**2)/n2)))
    if(z < 0):
     p = stats.norm.cdf(z)
    else:
     p = 1 - stats.norm.cdf(z)
     print (z,p)
```

In [7]:
```
Z_and_p(121,112,8,8,10,10)
```

```
2.5155764746872635 0.005941894621073 64
```

σ_1^2 and σ_2^2 Unknown, Assumed Equal

- Population means, independent samples
  - σ_1^2 and σ_1^2 known
  - σ_1^2 and σ_2^2 unknown
    - σ_1^2 and σ_1^2 assumed equal
    - σ_1^2 and σ_1^2 assumed unequal

Assumptions:

- Samples are randomly and independently drawn
- Populations are normally distributed
- Population variances are unknown but assumed equal

σ_1^2 and σ_2^2 Unknown, Assumed Equal

- The population variances are assumed equal, so use the two sample standard deviations and pool them to estimate σ
- use a t value with $(n_1 + n_2 - 2)$ degrees of freedom

Test Statistic, σ_1^2 and σ_2^2 Unknown, Equal

The test statistic for

$\mu_1 - \mu_2$ is:

$$t = \frac{\left(\bar{x}_1 - \bar{x}_2\right) - \left(\mu_1 - \mu_2\right)}{\sqrt{\frac{s_p^2}{n_1} + \frac{s_p^2}{n_2}}}$$

Where t has $(n_1 + n_2 - 2)$ d.f.,

and

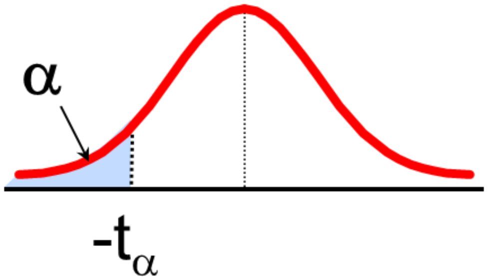
$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

Decision Rules

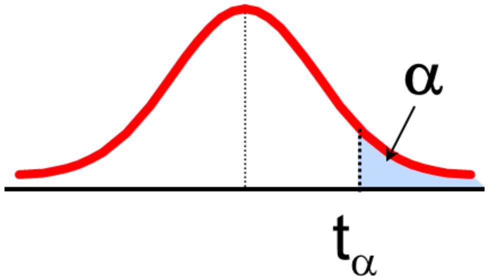
Two Population Means, Independent Samples, Variances Unknown

Lower-tail test:	Upper-tail test:	Two-tail test:
$H_0: \mu_1 - \mu_2 \geq 0$	$H_0: \mu_1 - \mu_2 \leq 0$	$H_0: \mu_1 - \mu_2 = 0$
$H_1: \mu_1 - \mu_2 < 0$	$H_1: \mu_1 - \mu_2 > 0$	$H_1: \mu_1 - \mu_2 \neq 0$

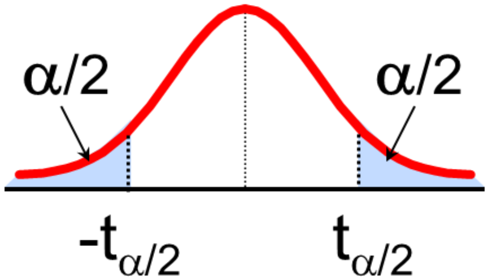
Decision Rules



Reject H_0 if
$t < -t_{(n1+n2-2),\ \alpha}$



Reject H_0 if
$t > t_{(n1+n2-2),\ \alpha}$



Reject H_0 if
$t < -t_{(n1+n2-2),\ \alpha/2}$ or
$t > t_{(n1+n2-2),\ \alpha/2}$

σ_1^2 and σ_2^2 Unknown, Assumed equal

- Two catalysts are being analyzed to determine how they affect the mean yield of a chemical process.
- Specifically, catalyst 1 is currently in use, but catalyst 2 is acceptable.
- Since catalyst 2 is cheaper, it should be adopted, providing it does not change the process yield.
- A test is run in the pilot plant and results in the data shown in table.
- Is there any difference between the mean yields?
- Use 0.05, and assume equal variances.

Observation Number	Catalyst 1	Catalyst 2
1	91.50	89.19
2	94.18	90.95
3	92.18	90.46
4	95.39	93.21
5	91.79	97.19
6	89.07	97.04
7	94.72	91.07
8	89.21	92.75

$\bar{x}_1 = 92.255$ $\bar{x}_1 = 92.733$

$S_1 = 2.39$ $S_2 = 2.98$

σ_1^2 and σ_2^2 Unknown, Assumed equal

1. The parameters of interest are μ_1 and μ_2, the mean process yield using catalysts 1 and 2, respectively, and we want to know if $\mu_1 - \mu_2 = 0$.
2. H_0: $\mu_1 - \mu_2 = 0$, or H_0: $\mu_1 = \mu_2$
3. H_1: $\mu_1 \neq \mu_2$
4. $\alpha = 0.05$
5. The test statistic is

$$t_0 = \frac{\bar{x}_1 - \bar{x}_2 - 0}{s_p \sqrt{\dfrac{1}{n_1} + \dfrac{1}{n_2}}}$$

σ_1^2 and σ_2^2 Unknown, Assumed equal

6. Reject H_0 if $t_0 > t_{0.025,14} = 2.145$ or if $t_0 < -t_{0.025,14} = -2.145$.

7. Computations: From Table 10-1 we have $\bar{x}_1 = 92.255$, $s_1 = 2.39$, $n_1 = 8$, $\bar{x}_2 = 92.733$, $s_2 = 2.98$, and $n_2 = 8$. Therefore

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2} = \frac{(7)(2.39)^2 + 7(2.98)^2}{8 + 8 - 2} = 7.30$$

$$s_p = \sqrt{7.30} = 2.70$$

and

$$t_0 = \frac{\bar{x}_1 - \bar{x}_2}{2.70\sqrt{\dfrac{1}{n_1} + \dfrac{1}{n_2}}} = \frac{92.255 - 92.733}{2.70\sqrt{\dfrac{1}{8} + \dfrac{1}{8}}} = -0.35$$

σ_1^2 and σ_2^2 Unknown, Assumed equal

8. Conclusions: Since $-2.145 < t_0 = -0.35 < 2.145$, the null hypothesis cannot be rejected. That is, at the 0.05 level of significance, we do not have strong evidence to conclude that catalyst 2 results in a mean yield that differs from the mean yield when catalyst 1 is used.

σ_1^2 and σ_2^2 Unknown, Assumed equal

```
In [12]: b =[ 89.19,90.95,90.46,93.21,97.19,97.04,91.07 , 92.75]

In [13]: a = [91.5, 94.18,92.18,95.39,91.79,89.07,94.72,89.21]

In [14]: stats.ttest_ind(a, b, equal_var = True)

Out[14]: Ttest_indResult(statistic=-0.3535908643461798, pvalue=0.7289136186068217)

In [21]: stats.t.ppf(0.025,14) #critical  t value

Out[21]: -2.1447866879169277
```

Thank You